

Physics Guided Rollout Restoration for Cyber Physical Smart Grids Coupled with Urban Electric Mobility Transition Based Closed Loop Restoration for Mobility Coupled Distribution Systems with Executable Actions

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Abstract—Urban distribution grids increasingly serve electric mobility loads whose charging demand, communication dependence, and traffic driven spatial movement change quickly during disruptions. A feeder disturbance can reduce communication service, weakened communication can disturb charging coordination. Restoration in mobility coupled distribution systems requires charging, communication, and delayed charging can feed back into mobility operation. We develop a physics guided rollout restoration method for cyber physical smart grids coupled with urban electric mobility. The controller estimates near future charging demand from graph structured mobility observations and evaluates candidate actions in a digital twin that represents feeder margin, charging backlog, communication service, mobility delay, and cross domain threat propagation. The central idea is a monotone propagation model and a risk calibrated restoration rule that allocates repair effort according to a future cascade envelope rather than the current outage map alone. A finite horizon risk bound is derived under explicit monotonicity and Lipschitz assumptions. Experiments on public city scale mobility records indicate improved stressed charging operation and cascading disruption response while normal operating efficiency is preserved. Parallel computation supports repeated closed loop evaluation over many disturbance realizations and operating states repair decisions that remain feasible after network and crew constraints are applied. This paper develops a physically constrained coupled rollout that evaluates finite horizon threat propagation through a calibrated central transition model. A requested decision passes through packet delivery, distribution system feasibility, station capacity, and integer crew routing before it can alter service or threat states. Measured electric vehicle charging records, public outage observations, and an independent distribution feeder support the empirical study. Paired comparisons show that central PC lowers operating cost relative to a forecast matched baseline that receives the same observations and demand forecasts but does not propagate threats. The improvement is statistically resolved under cascade and compound conditions and remains small during nominal operation. A worst set extension does

not improve mean or tail performance and is treated as a sensitivity variant. Electrical projection corrects infeasible charging requests when feeder limits bind. Packet delivery and integer repair completion also change the realized action. The analysis establishes finite horizon sensitivity for the continuous decision layer, a conditional regret bound for the finite route portfolio, and feasibility properties for the executed action. The resulting service restoration study connects measured charging demand with distribution systems, communication networks, and crew routing in one closed loop model.

Index Terms—Smart grid service restoration, electric mobility, digital twin, threat propagation, rollout decision making vehicle charging, distribution systems, communication networks, crew routing.

I. Introduction

Distribution grids are becoming coupled operating systems. Electric vehicles create charging demand whose location follows urban activity and road network access. Electric vehicles connect urban travel with distribution system operation. Charging demand consumes feeder margin and may create local backlog when service capacity is insufficient. Communication service supports charging coordination, sensing, and control. A feeder disturbance can therefore reduce communication service, increase charging backlog, and affect mobility operation before the operator sees the full spatial consequence. The same effect may move in the opposite direction when mobility congestion changes charging arrival patterns and raises the load on communication assisted charging services. This coupling changes restoration from a single layer repair task into a finite horizon decision problem. moves with daily activity, station access, road conditions, and driver choice. Its spatial pattern therefore differs from a conventional fixed load profile. Station planning and traffic power flow studies have shown that route choice and charging capacity can alter both travel and electrical states [1]–[3]. Operational pricing and navigation models extend this relation to real time charging decisions [4]–[6]. During a disruptive event, however, demand movement is only one part of the problem. Charging service also depends on feeder margin,

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communication availability, and the time at which a repair crew completes its work.

Power and transportation coupling has been studied from planning, pricing, and traffic assignment perspectives. Coordinated charging station planning and traffic power flow models describe how electric mobility reshapes distribution operation. More recent work has linked route choice, charging navigation, and grid states in coupled urban systems. These studies show that charging load cannot be treated as a static exogenous demand when the transportation layer is under stress. At the same time, cyber-physical resilience studies show that service disruption may spread across power, communication, and transportation layers rather than staying inside the first failed domain. A restoration method for such systems must therefore reason about both demand movement and threat propagation. These dependencies give restoration a temporal and cross domain structure. Communication supports sensing, charging coordination, and control delivery. Loss of local power may reduce that service after backup supply is exhausted. Delayed communication can then restrict charging actions and increase backlog. Road disruption can move demand toward stations whose electrical capacity is already scarce. Interdependent infrastructure studies explain why failures can spread across connected services [7], [8]. Distribution system resilience research further relates restoration decisions to service continuity under uncertain events [9], [10]. The resulting decision cannot be reduced to repairing the zone with the largest current loss because a moderate present threat may lie on a more consequential future path.

Forecasting alone is not enough for this restoration task. Graph forecasting models can estimate the near future distribution of mobility demand by using Demand forecasts provide useful information for this decision. Recent graph models represent spatial dependence and recent temporal patterns. Digital twin models can evaluate operating constraints and service loss in a form that is closer to the control objective. The missing link is an online decision rule that converts forecasted charging states and propagating cyber-physical threats into restoration actions. A greedy rule that repairs zones with the largest current loss may ignore zones that are not yet highly damaged but are positioned on a future cascade path. A neural rollout rule that follows predicted demand can also miss this effect if restoration is still tied to the current threat map. recurring temporal behavior in traffic and charging related series [11], [12]. Several models improve prediction under changing graph relations and limited observations [13]–[15]. Prediction error alone does not determine restoration quality. Two policies can receive the same forecast and still choose different actions because only one evaluates the propagation of road, power, and communication threats. Digital representations of physical operation provide a natural setting for this evaluation when their variables correspond to service and network constraints [16]–[18].

Existing work leaves three connected decisions

unresolved. The first is the conversion of a shared demand forecast into a finite horizon restoration score that includes action dependent threat propagation. The second is the conversion of continuous service priorities into actions that satisfy packet, feeder, station, and crew constraints. The third is the separation between a predictive repair priority and the integer completion event that actually changes the state. Treating a requested priority as completed repair would credit service before a crew arrives. Treating an unprojected charging request as served load would create the same inconsistency on the electrical side.

We develop a policy for cyber-physical smart grids coupled with urban electric mobility. The method is called physics-guided coupled. This paper develops physically constrained coupled rollout, abbreviated as PC rollout. A graph-recurrent network forecasts near future mobility driven charging demand. The forecast is passed to a digital twin whose equations represent charging service, backlog, communication capacity, feeder margin, and mobility delay. The digital twin is driven by a monotone threat propagation operator over road, power, and communication threat fields. Instead of allocating restoration effort to zones with the largest current loss, the policy allocates effort to zones with high future cascade exposure. This design keeps the learned predictor, the physical service equations, and the restoration rule separate, which makes the role of each part explicit in the analysis and in the experiments. compact graph recurrent model forecasts station arrivals and charging energy. A nonnegative transition model propagates mobility, power, and communication threats over a finite horizon. The primary central PC policy evaluates each candidate under the calibrated central transition. A separate robust extension evaluates a declared finite set of transition matrices. Requested charging and communication actions are reduced when packets miss their deadlines. Charging is then projected through an independent SMART-DS and OpenDSS calculation. Repair priorities form work orders, integer crew routes, and recorded completion events. Only executable service and completed repair enter the next realized state.

The paper makes the following theoretical and algorithmic contributions.

- It formulates cyber-physical smart grid operation with mobility driven charging demand as a finite horizon rollout problem in which the threat field is part of the state and evolves through a monotone cross domain propagation operator. an action dependent finite horizon restoration rule in which a calibrated transition model assigns marginal value to the first repair decision while the next observation closes the receding horizon calculation.
- It introduces a cascade envelope that summarizes future power, communication, and mobility threat exposure over the rollout horizon and converts this envelope into a concave restoration allocation rule with explicit feasibility and regularity properties. constructs an executable action algorithm

that distinguishes requested service, packet delivery, distribution system projection, station reassignment, dispatched work orders, and integer repair completion.

- It derives a finite horizon risk bound for both fixed action evaluation and the closed loop PC rollout policy. The bound relates rollout loss error to the spectral growth of the threat propagation operator, the action sensitivity of the digital twin, and the discount factor. It establishes finite horizon sensitivity for the continuous prediction and allocation layer, a conditional regret bound for a finite route portfolio, and feasibility and invariance results for executed actions.

The rest of this paper is organized as follows. Section II reviews the related literature. Section III gives the coupled decision model. Section IV presents the propagation model, the rollout policy, and the proof. Section V describes the implementation, experimental design, and empirical results. Section VI presents the system model, decision rule, and mathematical results. Section VII describes the empirical study and its results. Section VIII interprets the evidence and states data availability. Section IX concludes the paper.

II. Related Literature Work

A. Electric mobility and distribution grids

Electric mobility couples urban travel activity with distribution grid operation. Early charging deployment and demand models linked system operation. Charging deployment models connect station access, charging service, and grid impacts. These models are useful because they connect trip patterns with electricity demand, but most of them address planning or aggregate load estimation trip demand, and electrical capacity [1], [19], [20]. Later studies considered uncertainty in charging station planning, coordinated represent uncertain charging demand and coordinate investment across urban power and transportation networks, and extreme fast charging facilities systems [21], [22]. Extreme fast charging models add demanding electrical constraints to station location and service design [3]. Traffic power flow formulations further showed that transportation decisions and electrical constraints can be optimized place route decisions and distribution constraints in a common mathematical optimization model [2], [23]. This literature establishes the physical relation between mobility and charging load.

Recent smart grid studies have moved from planning toward operational coupling. Operational studies move the decision from capacity planning to traveler and station control. Pricing and navigation models coordinate electric vehicle coordinate route choice with charging and grid states. Distributed electric vehicle assignment and charging navigation have availability and distribution states [4]–[6]. Distributed assignment has also been studied in cyber physical settings when information and

control are exchanged across cyber physical components [24]. Robust planning for charging infrastructure has been used to improve resilience in coupled power distribution and transportation systems. Those studies motivate the coupled decision model used here. The difference is the decision time scale. Planning models decide where capacity should be built, while restoration must decide where scarce service and repair effort should move during an unfolding event. In this setting, feeder margin, communication service, charging backlog, and mobility delay must be evaluated together over a finite horizon. Resilience models add uncertain disruptions to the coupled power and transportation problem [10]. These models decide capacity, prices, or routes. Restoration adds a different time sequence because communication delivery, feeder feasibility, and crew completion determine when a requested action becomes effective.

The timing distinction changes the mathematical role of an action. In a planning model, installed capacity is available once the investment decision is made. In a navigation model, a recommended route can enter the traffic state within the next assignment interval. A restoration request instead passes through message delivery, travel, and service before it changes local conditions. Charging coordination studies provide the demand and network relations needed for this setting [4], [24]. They leave room for an event based state update that records service only after the corresponding electrical or crew condition has been met.

B. Cyber physical resilience and digital twins

Interdependent infrastructure theory has long recognized that failures can move describes failure transfer across network layers. Foundational studies described infrastructure interdependencies and reviewed simulation models for interdependent critical systems and the resulting service consequences [7], [8]. Network science then showed that coupled networks may experience abrupt cascade effects models show that coupled systems can develop abrupt cascades [25]. Spreading process analysis provides mathematical tools for bounding the growth of such processes supplies tools for studying the growth and control of such dynamics on graphs [26]. Power system resilience studies connect these ideas to restoration, service continuity, and operation under disruptive events. research connects these ideas with preventive operation, restoration, and service continuity [9], [10], [27]. Reinforcement learning has also been used to represent sequential restoration decisions [28]. The present model uses this temporal view but keeps the action mechanism explicit so that a repair priority and a completed repair remain different states.

Physically constrained security research considers whether learned grid functions and adversarial inputs satisfy power balance, operating limits, and stealth requirements. Formal evaluation has been developed for intelligent security assessment and tree ensemble

applications [29], [30]. Hybrid learned and classical load frequency control has also been studied under physically admissible attack models [31]. Those studies constrain cyber security analysis with grid equations. The restoration problem here instead allocates charging, communication, and repair resources over time, then subjects each request to physical and operational execution rules.

Digital twins provide a modeling basis for cyber physical decision support. General digital twin representations provide a related basis for testing a decision before it is applied. General studies define the connection between among a physical asset, a virtual its computational representation, and a data exchange process. Work in networked sensing and industrial informatics emphasizes that the digital representation should remain operationally meaningful rather than only visually descriptive. In the present setting, a useful twin must represent the variables that constrain grid action. These variables include data exchange [16], [17]. Networked and industrial models stress the need for an operational relation between the representation and the asset [18], [32]. The model in this paper represents charging service, backlog, feeder margin, communication capacity, and restoration resources, threat propagation, feeder feasibility, and crew progress. These variables are used to calculate an action and its realized consequences rather than to create a visual copy of the system.

This operational use requires consistent clocks across the represented services. Demand and threat are observed at the decision interval. Packet events occur within that interval. Crew travel and repair can continue across several intervals. A continuous simulator can rank candidate actions, while the realized state must wait for the discrete event that makes the action effective. Digital representation studies support the connection between physical state and computational state [17], [18]. The present formulation uses that connection to separate the predictive state from the executed state and to keep their update rules explicit.

The restoration problem considered here draws from both lines of work. The threat state is not only an outage indicator. It is a vector field over mobility, power, and communication domains, and it evolves through spatial spread and cross domain couplings. This representation follows the cascade view from interdependent infrastructure analysis, while the cost evaluation uses service variables that can be interpreted by grid operators. The digital twin is therefore not used as a visualization device. It acts as the finite horizon simulator in which restoration actions are tested before the first action is applied.

C. Graph forecasting for charging states

The distribution grid observes future electric mobility demand. Charging load is observed through forecasts because charging load is driven by travel activity. Diffusion

convolutional recurrent networks and spatiotemporal graph convolutional networks established influential approaches for modeling directed traffic dependence. Graph WaveNet and adaptive graph recurrent models improved spatial adaptation and temporal receptive fields. Recent normalization, embedding, and meta graph methods show that graph learning remains effective for demand and flow prediction. Traffic prediction surveys and time series studies provide additional evidence for graph based state estimation. Recent studies on fast spatiotemporal learning, self supervised graph learning, and few sample prediction further support the use of graph based state estimators in urban demand data. Automated dilated graph modeling gives related evidence for traffic state prediction. A graph recurrent model is used here because the main scientific question is the conversion of forecasted charging states and propagating threats into restoration decisions. Travel activity determines when and where vehicles request service. Reviews of graph forecasting describe the main spatial and temporal structures used for traffic prediction [12], [33]. Recent models adapt graph relations, separate temporal scales, and transfer information to sparsely observed locations [11], [13], [34]. Few observation and self supervised formulations further address limited target data [14], [35]. Physics informed graph models offer another route for connecting network structure with prediction [15]. Direct measurement studies nevertheless show that charging behavior retains substantial spatial and temporal variation [36].

These forecasting methods mainly. Most forecasting models optimize prediction error. Restoration adds a different requirement. Prediction errors matter through their effect on action selection, and a forecast that is accurate on average can still be insufficient when a future cascade changes the value of restoration at particular zones. For this reason, the predictor, whereas restoration depends on the decision induced by a forecast. A forecast can have low average error and still assign insufficient attention to a zone on a future propagation path. The forecaster in this paper is intentionally kept as a state estimator rather than treated as the main decision maker. The rollout policy uses the forecast to estimate future charging load, then uses the propagation model to form the cascade envelope that drives restoration. This distinction separates demand estimation from cyber physical risk allocation, which is central to the method and to the experimental comparisons. therefore a shared state estimator. Central PC and the forecast matched baseline receive the same forecast, current threat, backlog, and realized history. Their difference is the propagation calculation used to value a candidate route. This comparison isolates the transition model from demand prediction and preserves equal information at the decision time.

Public infrastructure data also have distinct evidentiary roles. Measured charging transactions provide station demand. County outage observations provide temporal

information about power events [37]. An independent distribution feeder determines whether a charging request is electrically feasible. Keeping these roles separate avoids representing records from different systems as one geographically coincident network. The method joins them through stated model variables and tests the resulting action through packet, feeder, station, and crew constraints.

The literature therefore supplies four strands of the present problem. Charging studies describe demand movement and service capacity. Infrastructure resilience studies describe propagation across connected domains. Graph forecasting estimates the near future demand state. Digital representations connect a candidate decision to operational constraints. The algorithm in the next section combines these strands through a common information set and an event based execution rule. This combination permits a direct comparison between transition evaluation and a baseline that receives the same forecast without using the transition model.

III. Coupled Smart Grid Operation System Model and Restoration Method

A. Operating state

Let $G = (V, E)$ be an urban charging service graph with N zones. The weighted adjacency matrix W is obtained from origin and destination flows that describe where electric mobility demand is likely to move. A symmetrized normalized matrix is denoted by A . At decision time t , the measured and estimated state is

$$x_t = (m_t, e_t, q_t, c_t, p_t, z_t), \quad (1)$$

where $m_t \in \mathbb{R}_+^N$ is mobility activity that drives charging demand, $e_t \in \mathbb{R}_+^N$ is charging energy demand, $q_t \in \mathbb{R}_+^N$ is charging backlog, $c_t \in \mathbb{R}_+^N$ is communication load, $p_t \in \mathbb{R}_+^N$ is feeder load, and $z_t = (z_t^r, z_t^p, z_t^c)$ is the mobility, power, and communication threat field. Each threat component field lies in $[0, 1]^N$. Larger values indicate less available service in that domain. It is a normalized service degradation intensity, not the binary state of one breaker, line, or router. For example, $z_i^p = 0.6$ means a 60% derating in the zone level power dependent service factor. The value $z_i^r = 0.6$ produces a 60% proportional mobility impairment before the communication term. A binary outage can be represented as a high intensity limiting case. The scenario scope includes cyber induced loss of sensing and control availability and physical or service failures. The state describes service degradation without assigning a particular attack type.

The action-requested decision is

$$a_t = (u_t^{\text{ch}}, u_t^{\text{com}}, u_t^{\text{res}}). \quad (2)$$

The first component term allocates charging service capacity, the second allocates communication service capacity, and the third allocates restoration effort in the coupled cyber-physical grid. is a continuous restoration priority

used by the predictive scoring layer. It is not itself a completed repair. The feasible set is

$$\begin{aligned} \mathcal{A} = \{a_t \geq 0 \mid & \mathbf{1}^\top u_t^{\text{ch}} \leq B_{\text{ch}}, \\ & \mathbf{1}^\top u_t^{\text{com}} \leq B_{\text{com}}, \\ & \mathbf{1}^\top u_t^{\text{res}} \leq B_{\text{res}}\}. \end{aligned} \quad (3)$$

Here B_{ch} , B_{com} , and B_{res} are the available charging, communication, and restoration budgets normalized restoration resources during one decision interval. They are scalar system resources, while the corresponding action vectors describe their spatial allocation over zones. The continuous restoration variable is a service level priority used to form candidate work orders. Section IV converts these candidates into integer crews, explicit road routes, service times, and repair completion events. At every policy hour, charging action is also projected through the SMART-DS and OpenDSS feasibility loop before service is credited and backlog is updated.

B. Grid and service equations

The digital twin uses a set of differentiable and continuous prediction layer uses monotone service equations that are continuous, piecewise smooth, and Lipschitz on the stated bounded domain. The minimum, positive part, and clipping operators need not be differentiable at their switching points. For zone i and rollout step h , available communication service is

$$\bar{c}_{i,h} = u_{i,h}^{\text{com}}(1 - z_{i,h}^p)(1 - \alpha_c z_{i,h}^c), \quad (4)$$

where $\alpha_c \in [0, 1]$ measures direct communication degradation. The load placed on communication is

$$c_{i,h} = \beta_m m_{i,h} + \beta_e e_{i,h} + \beta_r z_{i,h}^r + \beta_p z_{i,h}^p. \quad (5)$$

The resulting communication loss is

$$\ell_{i,h}^c = [c_{i,h} - \bar{c}_{i,h}]_+. \quad (6)$$

Communication support affects charging coordination and mobility service through

$$\eta_{i,h}^c = 1 - \min \left\{ 0.9, \frac{\ell_{i,h}^c}{1 + c_{i,h}} \right\}. \quad (7)$$

This algebraic relation is an hourly workload model. Executability is supplied by a finite buffer packet emulator with retransmission and deadline logic. If $d_{i,h} \in [0, 1]$ is the simulated fraction of control packets delivered before their deadline, the automated action applied to the service and threat transitions is

$$\tilde{a}_{i,h} = d_{i,h} a_{i,h}. \quad (8)$$

The emulator keeps full service rate while backup power is available and applies power induced service derating only after backup exhaustion. Packet loss and delay therefore change the implemented action rather than appearing only as a later diagnostic. Before charging service is credited, the packet reduced request is mapped to three

phase SMART-DS loads and scaled by the largest feasible fraction $\alpha_h^{\text{AC}} \in [0, 1]$ found by bisection,

$$u_{i,h}^{\text{ch,exec}} = \alpha_h^{\text{AC}} \tilde{u}_{i,h}^{\text{ch}}. \quad (9)$$

The feeder must converge, all energized node voltages must lie in $[0.95, 1.05]$ p.u., and maximum line loading must not exceed 1.0 p.u. Charging service is therefore

$$s_{i,h}^{\text{ch}} = \min \left\{ e_{i,h} + \rho_q q_{i,h}, u_{i,h}^{\text{ch,exec}} (1 - z_{i,h}^p) \eta_{i,h}^c \right\}. \quad (10)$$

The charging backlog evolves as

$$q_{i,h+1} = [e_{i,h} + \rho_q q_{i,h} - s_{i,h}^{\text{ch}}]_+. \quad (11)$$

~~The feeder margin violation is~~ We impose $0 \leq \rho_q < 1$ and set $\rho_q = 0.22$. If hourly arrival energy is bounded by $\bar{e}$, then $q_{i,h+1} \leq \bar{e} + \rho_q q_{i,h}$ and consequently $q_{i,h} \leq \max\{q_{i,0}, \bar{e}/(1 - \rho_q)\}$.

The original zone level feeder margin surrogate is retained only as an interpretable service diagnostic,

$$\ell_{i,h}^{\text{fm}} = \left[p_i^0 + s_{i,h}^{\text{ch}} - \bar{p}_i (1 - \alpha_p z_{i,h}^p) \right]_+, \quad (12)$$

where p_i^0 is ~~an abstract~~ base load and $\bar{p}_i$ is ~~the local feeder capacity, an abstract local capacity~~. Executable charging is determined by the online OpenDSS projection in (9). Mobility delay is

$$\ell_{i,h}^m = m_{i,h} (z_{i,h}^r + \rho_c (1 - \eta_{i,h}^c)). \quad (13)$$

Unservd charging is defined once as $\ell_{i,h}^e = [e_{i,h} + \rho_q q_{i,h} - s_{i,h}^{\text{ch}}]_+$, and power dependent service loss is $\ell_{i,h}^p = z_{i,h}^p (e_{i,h} + \rho_q q_{i,h})$. The coefficients in these equations have fixed physical roles. The parameters β_m , β_e , β_r , and β_p convert mobility activity, charging demand, mobility threat, and power threat into communication workload. The coefficient ρ_q is backlog carryover, ~~ρ_u is direct charging relief~~, and α_p is ~~feeder the abstract~~ capacity derating caused by power threat. The ~~coefficients coefficient~~ ρ_c and ρ_e ~~convert converts~~ communication service reduction and ~~unmet charging demand into mobility delay~~ into delayed trip equivalents. The stage risk is

$$L(x_h, a_h) = \sum_{i=1}^N (\lambda_m \ell_{i,h}^m + \lambda_e \ell_{i,h}^e + \lambda_c \ell_{i,h}^c + \lambda_p \ell_{i,h}^p + \lambda_z [z_{i,h}^p z_{i,h}^c + z_{i,h}^r (1 - \eta_{i,h}^c)]). \quad (14)$$

The nonnegative weights λ_m , λ_e , λ_c , λ_p , and λ_z set the relative ~~units contributions~~ of mobility delay, unmet charging demand, communication loss, ~~feeder margin violation, and simultaneous power dependent service loss~~, and cross domain exposure. The objective is dimensionless because these heterogeneous terms are combined with fixed relative weights. It is not a monetary operating cost. Unserved charging appears only through ℓ^e . All coefficients are held fixed across policies.

For a feasible policy π , the finite horizon rollout objective at decision time t is

$$J_H^\pi(x_t, \zeta_t) = \sum_{h=0}^{H-1} \gamma^h L(x_{t+h}, a_{t+h}), \quad (15)$$

$$a_{t+h} = \pi(x_{t+h}, \zeta_{t+h}, A). \quad (16)$$

The ~~objective in (15) is dimensionless because its heterogeneous service terms are combined with fixed relative weights~~. The closed loop controller applies only the first action from the horizon calculation. At the next decision time, the observed state replaces the previous prediction and the calculation is repeated. In this receding horizon interpretation, the next state measurement absorbs forecast errors and threat disturbances before the following decision.

C. ~~Forecasting component~~ Demand forecasting

The forecaster estimates mobility and charging demand.

$$\hat{y}_{t+1, \dots, t+H} = f_\theta(y_{t-L+1}, \dots, y_t, A), \quad (17)$$

where $y_t = (m_t, e_t)$, L is the historical window, and H is the rollout horizon. The implemented model first mixes node features through A , then updates zone states using a recurrent cell, and finally maps the hidden state to the horizon. Keeping the forecaster small makes its role clear inside the larger decision model. Classical baselines and the learned model are compared in Section ~~V-IV~~ so that the decision results can be interpreted relative to the forecasting quality.

The input series is transformed by a logarithmic map and normalized using only the training portion of the calendar year. For a time index τ ,

$$\tilde{y}_\tau = \frac{\log(1 + y_\tau) - \mu}{\sigma}. \quad (18)$$

The graph recurrent forecaster uses one graph mixing operation before each recurrent update. With hidden state h_ℓ inside the historical window,

$$g_\ell = A \tilde{y}_{t-L+\ell}, \quad (19)$$

$$h_\ell = \text{GRU}(W_g g_\ell, A h_{\ell-1}), \quad (20)$$

$$\hat{y}_{t+1, \dots, t+H} = W_o \phi(h_L). \quad (21)$$

Here ϕ denotes the output transformation used by the prediction head. The ~~training loss combines a robust forecast error with two physical regularizers~~. ~~primary forecaster is trained with the Huber loss~~

$$\mathcal{L}_{\text{for}} = \mathcal{L}_{\text{Huber}}(\hat{y}, y). \quad (22)$$

~~The coefficient α_i is Its spatial inductive bias comes from the fixed origin and destination graph mixing inside the recurrent model, not from an additional physical penalty. Three independent training runs are performed, and the energy per activity ratio for zone i estimated from the training period, σ_E is the training scale of charging energy, and λ_E and λ_A weight the two regularizers. The energy term reflects the empirical relation between trip~~

activity and charging demand, while the graph smoothness term discourages isolated spikes that are inconsistent with adjacent high flow zones. These terms supplement the data fit and keep the learned forecast compatible with the digital twin variables that later enter (14) model retained for the decision study is selected only by its validation score. An auxiliary graph smooth training mode produced a higher validation error and was excluded before the restoration policies were compared.

IV. Physics Guided Smart Grid Rollout

A. Transition model and action construction

B. Threat propagation

Let $\zeta_h = (z_h^r, z_h^p, z_h^c) \in [0, 1]^{3N}$. The propagation model is Two transitions are kept explicit because prediction and execution have different roles. The continuous transition used only to score a requested restoration priority is

$$\hat{\zeta}_{h+1} = \Pi_Z \left(M\hat{\zeta}_h - Ru_h^{\text{pri}} + \hat{\xi}_h \right), \quad (23)$$

where $\mathcal{Z} = [0, 1]^{3N}$, Π_Z is Euclidean projection, $M \geq 0$ is the cross domain propagation matrix, Rr_h represents restoration effects derived from the restoration action, Dd_h represents stress created by backlog, congestion, and service loss, and ξ_h is an exogenous disturbance. The matrix M contains spatial spreadthrough $R > 0$ maps the zone level priority u_h^{pri} to predicted road, power, and communication relief, and ξ_h is the innovation assumed by the scorer. Equation (23) does not update the realized state.

For execution, let $\bar{r}_{i,h} \in \{0, 1\}$ indicate that a dispatched work order in zone i has completed by the end of interval h , and define $D(\bar{r}_h) = I_3 \otimes \text{diag}(1 - \bar{r}_h)$. The main simulator uses

$$\zeta_{h+1} = D(\bar{r}_h)\Pi_Z (M\zeta_h + \xi_h). \quad (24)$$

Waiting, travel, and service do not change $\bar{r}_h$. Only a recorded crew completion event changes it from zero to one. Thus no continuous priority is credited as realized repair. With $K_d = \rho_d I + \sigma A$ for $d \in \{r, p, c\}$, the implemented matrix is

$$M = \begin{bmatrix} K_r & \tau_{pr}I & \tau_{cr}I \\ 0 & K_p & \tau_{cp}A \\ \tau_{rc}A & \tau_{pc}I & K_c \end{bmatrix}. \quad (25)$$

The block structure in (25) separates persistence, spatial spread, and directed cross domain effects. The normalized adjacency A enters both within domain spread and selected cross domain paths. Boulder charging deficits estimate ρ_r and cross domain couplings from σ . EAGLE-I Boulder County outages estimate ρ_p . Aligned outage and charging records bound τ_{pr} . The packet emulator constructs the power to communication interval for τ_{pc} . Coefficients lacking joint field observations are assigned sensitivity ranges. Bootstrap and sign constrained intervals define an uncertainty set $\mathcal{U}$. The central matrix has estimated spectral radius 1.028, so no asymptotic contraction is claimed. Projection makes the implemented

state set invariant and the policy is evaluated over a finite six hour horizon and a 100, from communication to mobility operation, and from mobility congestion to charging stress. Restoration has the opposite sign inside Rr_h , but the projected map remains monotone in the unrepaired threat field for fixed actions.

The future cascade envelope used by PC rollout is 000 step stress test.

PC rollout scores restoration by its action dependent marginal benefit. Let $w_{i,h}$ be a bounded exposure weight formed from the shared mobility and energy forecast, and let $\Omega = \text{diag}(\omega_r, \omega_p, \omega_c)$. For a candidate first restoration vector v , define

$$\mathcal{C}_t(v) = \sum_{h=0}^{H-1} \sum_{i=1}^N w_{i,h} \mathbf{1}^\top \Omega \hat{\zeta}_{i,t+h}(v), \quad (26)$$

The first term measures local cumulative exposure and the second term measures exposure arriving from adjacent high flow zones. The where $\hat{\zeta}_{i,t+h}(v)$ follows (23), using v only in the first predicted transition. For reference quantum δ , the marginal benefit of repairing zone i is $b_{i,t}(M) = [\mathcal{C}_t(0, M) - \mathcal{C}_t(\delta e_i, M)]_+$. The primary policy uses the calibrated central matrix M_c , $b_{i,t}^{\text{cen}} = b_{i,t}(M_c)$. A separately evaluated robust extension uses the lower envelope

$$b_{i,t}^{\text{rob}} = \min_{M \in \mathcal{U}_V} b_{i,t}(M), \quad (27)$$

where $\mathcal{U}_V$ contains the central matrix, joint lower and upper vertices, and the endpoints of each individual coefficient (21 matrices in the tested set). The policy specific time indexed restoration score is

$$s_{i,t}^{\text{res}} = \omega_p z_{i,t}^p + \omega_c z_{i,t}^c + \omega_r z_{i,t}^r + \omega_\Gamma b_{i,t}^\pi. \quad (28)$$

Here $b^\pi = b^{\text{cen}}$ for the primary central PC method and $b^\pi = b^{\text{rob}}$ only for the robust extension. The restoration action uses a small uniform reserve and a concave risk allocation.

$$u_i^{\text{res}} = \frac{\eta B_{\text{res}}}{N} + (1 - \eta) B_{\text{res}} \frac{\sqrt{s_i^{\text{res}} + \epsilon}}{\sum_{j=1}^N \sqrt{s_j^{\text{res}} + \epsilon}}, \quad (29)$$

where η is the reserve fraction. The square root creates concavity. It prevents a single zone from taking nearly all restoration effort while still giving more effort to zones with larger cascade exposure. The weights $\omega_A, \omega_p, \omega_c, \omega_r$, and ω_Γ are nonnegative constants that control adjacency exposure and domain specific exposure in the restoration score direct domain exposure and the multistep action benefit. The constant ϵ is a small positive stabilizer that keeps every zone eligible for a nonzero reserve.

Charging and communication capacities are assigned with the same allocator so that all resource decisions share the same feasibility structure. For any budget B , nonnegative score vector v , and reserve fraction η_B , define

$$\Phi_i(B, v, \eta_B) = \frac{\eta_B B}{N} + (1 - \eta_B) B \frac{\sqrt{v_i + \epsilon}}{\sum_{j=1}^N \sqrt{v_j + \epsilon}}. \quad (30)$$

PC rollout uses

$$u_i^{\text{ch}} = \Phi_i(B_{\text{ch}}, g^{\text{ch}}, \eta_{\text{ch}}), \quad (31)$$

$$u_i^{\text{com}} = \Phi_i(B_{\text{com}}, g^{\text{com}}, \eta_{\text{com}}), \quad (32)$$

where

$$g_{i,t}^{\text{ch}} = \sum_{h=1}^H \hat{e}_{i,t+h} + \kappa_q q_{i,t} + \kappa_{\text{ch}} (z_{i,t}^r + z_{i,t}^p + z_{i,t}^c), \quad (33)$$

$$g_{i,t}^{\text{com}} = \sum_{h=1}^H \hat{m}_{i,t+h} + \kappa_E \left(\sum_{h=1}^H \hat{e}_{i,t+h} + \kappa_q q_{i,t} \right) + \kappa_{\text{com}} (z_{i,t}^r + z_{i,t}^p + z_{i,t}^c). \quad (34)$$

Service capacity follows predicted demand, while ~~restoration~~ the requested restoration priority follows the future cascade envelope. Charging and communication resources serve expected load. ~~Restoration resources reduce the propagation terms that drive later losses. The coefficients~~ In the scoring layer, priority perturbs (23). In the execution layer, only the completion indicator in (24) changes realized threat. The coefficients κ_q , κ_{ch} , κ_E , and κ_{com} determine ~~the extent to which current threat~~ how observed backlog, current threat, and forecasted charging demand alter the ~~service capacity scores~~ time varying service scores. For the main experiment, an initial threat realization and adjacency based propagation create a policy independent set of at most 36 work orders. Four integer crews start from a common depot. Candidate routes are constructed from static, greedy, forecast matched, central PC, robust PC, and uniform priorities using the same jobs, travel matrix, service time draws, and dispatch uniforms for every policy. Forecast matched, central PC, and robust PC see exactly this same finite route portfolio. Forecast matched minimizes current exposure weighted completion without propagating M . Central PC minimizes predicted integrated risk under the calibrated central M . Robust PC minimizes the worst predicted integrated risk over the uncertainty set of 21 matrices. Travel uses geocoded station locations, 1.25 road circuitry, and 30 km/h. service time scenarios are informed by observed EAGLE-I post peak recovery times. Because those county traces are not work orders, they are an explicit scenario prior rather than measured crew durations. A dispatched job changes no restoration state while waiting or being serviced. Only its logged completion event clears local threat, after which the zone remains repaired within the horizon. The mobility graph A is used for propagation. Crew travel uses the separate geocoded road distance matrix.

Algorithm 1 states the decision rule. Forecasting, propagation, service evaluation, and restoration allocation are kept as separate calculations so that the effect of each ~~component~~ decision term can be inspected.

Lemma 1. The restoration ~~action~~ priority allocation in (29) is feasible for every nonnegative score vector.

Algorithm 1: ~~Physics-guided~~ Physically constrained coupled rollout

Require: Current state x_t , graph A , budgets B_{ch} , B_{com} , B_{res} , horizon H , and trained forecaster f_θ

Ensure: ~~First feasible restoration and service action~~ α_t Executable service action and event driven repair update

- 1: Predict ~~$\hat{y}_{t+1:t+H}$~~ $\hat{y}_{t+1:t+H}$ with f_θ
- 2: Initialize mobility, power, and communication threats from the observed state
- 3: for each zone i do
- 4: ~~Propagate threats with (23) over the horizon~~ Roll out (23) at M_c with no first action and candidate δe_i
- 5: ~~Form the cascade envelope with (26)~~ Set $b_{i,t}^{\text{cen}} \leftarrow [C_t(0, M_c) - C_t(\delta e_i, M_c)]_+$
- 6: ~~Optionally evaluate $b_{i,t}^{\text{rob}}$ over $\mathcal{U}_i$ for the robust extension~~
- 7: end for
- 8: Allocate charging and communication service with (30)
- 9: ~~Allocate restoration~~ Compute requested restoration priority u_t^{pri} with (29)
- 10: ~~Evaluate the digital twin loss in (14)~~ Translate priorities into requested work orders
- 11: Apply the ~~first action and repeat after the next observation~~ packet deadline gate to form dispatched work orders
- 12: ~~Construct the common finite route portfolio over dispatched orders~~
- 13: ~~Select a route by current, central M , or worst set scoring according to policy~~
- 14: ~~Gate charging and communication actions by packets delivered on time~~
- 15: ~~Project charging through OpenDSS and credit only the executable service~~
- 16: ~~Advance (24) only when a crew completion event occurs~~
- 17: ~~Observe the next state and repeat~~

Proof. Each score satisfies $s_i^{\text{res}} \geq 0$ and $\epsilon > 0$, so every square root in (29) is positive. Hence $u_i^{\text{res}} \geq 0$ for all zones. Summing (29) over all zones gives

$$\begin{aligned} \sum_{i=1}^N u_i^{\text{res}} &= \sum_{i=1}^N \frac{\eta B_{\text{res}}}{N} + (1 - \eta) B_{\text{res}} \frac{\sum_{i=1}^N \sqrt{s_i^{\text{res}} + \epsilon}}{\sum_{j=1}^N \sqrt{s_j^{\text{res}} + \epsilon}} \\ &= \eta B_{\text{res}} + (1 - \eta) B_{\text{res}} = B_{\text{res}}. \end{aligned} \quad (35)$$

The ~~action~~ requested priority satisfies the nonnegativity and budget constraints. $\square$

B. Mathematical properties

The Lipschitz results below concern the continuous prediction and allocation operators before execution. They exclude packet deadline switches, OpenDSS acceptance tests and bisection, station flow active set changes,

finite route argmin selection, and integer completion events. Those executable operations are covered instead by the finite portfolio bound and constraint preservation proposition below.

The analysis uses the Euclidean norm. The projection onto a closed convex set is nonexpansive, so

$$\|\Pi_{\mathcal{Z}}(u) - \Pi_{\mathcal{Z}}(v)\| \leq \|u - v\|. \quad (36)$$

~~Assumption-Condition 1. For all feasible~~ On the stated bounded continuous prediction domain and for all feasible requested actions, the ~~stress map~~ $d(x, a)$ is Lipschitz in x with constant L_d , and the one step physical state transition $F(x, a, \zeta)$ ~~one step service state operator~~ $F_c(x, a, \zeta)$ is Lipschitz in (x, ζ) with finite constants L_x and L_z .

$$\|F_c(x, a, \zeta) - F_c(\tilde{x}, a, \tilde{\zeta})\| \leq L_x \|x - \tilde{x}\| + L_z \|\zeta - \tilde{\zeta}\|. \quad (37)$$

~~Assumption-Condition 2. The stage risk in (14) is~~ Lipschitz in (x, ζ) with constants C_x and C_z for all feasible actions.

$$|L(x, a, \zeta) - L(\tilde{x}, a, \tilde{\zeta})| \leq C_x \|x - \tilde{x}\| + C_z \|\zeta - \tilde{\zeta}\|. \quad (38)$$

~~The assumption holds for the implemented model because every service equation is a composition of affine maps, minimum operators, positive part operators, and bounded products on the compact feasible set. Lemma 2. Conditions 1 and 2 hold for the continuous service equations on the stated bounded domain. The continuous state transition and stage risk are also Lipschitz in the feasible action argument.~~

~~Proof. Let $\mathcal{D}$ denote the product of the closed state, threat, and feasible action intervals. The backlog bound in Section III gives a finite upper bound for q because $0 \leq \rho_q < 1$ and arrivals are bounded. All remaining state variables have finite upper bounds by the definition of $\mathcal{D}$. The scalar maps $r \mapsto [r]_+$ and $r \mapsto \min\{c, r\}$ are nonexpansive. Euclidean projection and componentwise clipping are also nonexpansive. Every affine map has a finite operator norm.~~

~~For bounded scalars a, a', b, b' , the product satisfies~~

$$|ab - a'b'| \leq |a| |b - b'| + |b'| |a - a'|. \quad (39)$$

~~The right hand side has finite coefficients on $\mathcal{D}$. The only quotient in the service equations has the form $g(\ell, c) = \ell / (1 + c)$. If $0 \leq \ell, \ell' \leq \bar{\ell}$ and $c, c' > 0$, then~~

$$|g(\ell, c) - g(\ell', c')| \leq |\ell - \ell'| + \bar{\ell} |c - c'|. \quad (40)$$

~~This follows by adding and subtracting $\ell' / (1 + c)$ and using $(1 + c)^{-1} \leq 1$ together with $|(1 + c)^{-1} - (1 + c')^{-1}| \leq |c - c'|$. Consequently, every service term in (4) through (15) is Lipschitz on $\mathcal{D}$. A finite sum and a finite composition of these terms are Lipschitz. The continuous state update therefore admits finite constants L_x , L_z , and L_a . Applying the same product bound to the cross domain terms in (14), followed by the triangle inequality over the finite set of zones, gives finite constants C_x , C_z , and C_a . This proves Conditions 1 and 2. $\square$~~

Lemma 2.3. For fixed action sequence and disturbance sequence, if two rollout trajectories start from threat fields ζ_0 and $\tilde{\zeta}_0$ and from the same physical state, then

$$\|\zeta_h - \tilde{\zeta}_h\| \leq \|M\|^h \|\zeta_0 - \tilde{\zeta}_0\|. \quad (41)$$

~~Proof. Let $\Delta_h = \|\zeta_h - \tilde{\zeta}_h\|$ and $E_h = \|x_h - \tilde{x}_h\|$. From (23) and (36), Applying this recursion once gives $\Delta_1 \leq \|M\|\Delta_0 + \|D\|L_d E_0$. Applying it repeatedly yields (41) by induction. The base case is immediate. If the statement holds for h , substitution into the one step recursion gives the statement for $h + 1$. This completes the proof. Because the action and innovation sequences are shared, (23) and the nonexpansiveness in (36) give $\Delta_{h+1} \leq \|M\|\Delta_h$. Iteration proves (41). $\square$~~

Lemma 3.4. Under ~~Assumption-Condition 1~~, the ~~physical service~~ state error satisfies

$$E_h \leq \sum_{k=0}^{h-1} L_x^{h-1-k} L_z \Delta_k. \quad (42)$$

~~Proof. The two rollouts use the same action sequence. Assumption-Condition 1 gives $E_{h+1} \leq L_x E_h + L_z \Delta_h$. Since the initial physical state is the same, $E_0 = 0$. Expanding the recursion gives $E_1 \leq L_z \Delta_0$ and $E_2 \leq L_x L_z \Delta_0 + L_z \Delta_1$. Repeating the expansion gives (42). $\square$~~

Theorem 1. Let $J_H(x_0, \zeta_0)$ be the discounted H step risk of a fixed feasible action sequence, and let $\tilde{J}_H(x_0, \tilde{\zeta}_0)$ be the corresponding risk for an initial threat estimate $\tilde{\zeta}_0$. For any $\alpha > 0$, define

$$\chi_\alpha = \max \{\|M\| + \alpha L_z, L_x\}. \quad (43)$$

If $\|\zeta_0 - \tilde{\zeta}_0\| \leq \varepsilon$, then

$$|J_H(x_0, \zeta_0) - \tilde{J}_H(x_0, \tilde{\zeta}_0)| \leq \varepsilon \left(C_z + \frac{C_x}{\alpha} \right) \sum_{h=0}^{H-1} (\gamma \chi_\alpha)^h. \quad (44)$$

~~When For every finite H the sum is finite. If $\gamma \chi_\alpha < 1$, the right hand side is bounded by it is also uniformly bounded as the horizon increases by~~

$$\varepsilon \left(C_z + \frac{C_x}{\alpha} \right) \frac{1}{1 - \gamma \chi_\alpha}. \quad (45)$$

~~Proof. Let $\Delta_h = \|\zeta_h - \tilde{\zeta}_h\|$ and $E_h = \|x_h - \tilde{x}_h\|$. Lemma 2 and Assumption-3 and Condition 1 give the one step bounds~~

$$\Delta_{h+1} \leq \|M\| \Delta_h, \quad (46)$$

$$E_{h+1} \leq L_z \Delta_h + L_x E_h. \quad (47)$$

For a fixed $\alpha > 0$, define the weighted error $V_h = \Delta_h + \alpha E_h$. Multiplying (47) by α and adding (46) gives

$$\begin{aligned} V_{h+1} &\leq (\|M\| + \alpha L_z) \Delta_h + \alpha L_x E_h \\ &\leq \chi_\alpha \Delta_h + \alpha \chi_\alpha E_h = \chi_\alpha V_h. \end{aligned} \quad (48)$$

The two rollouts start from the same physical state, so $E_0 = 0$ and $V_0 = \Delta_0 \leq \varepsilon$. Induction gives $V_h \leq \chi_\alpha^h \varepsilon$

for every h . Since $\Delta_h \leq V_h$ and $E_h \leq V_h/\alpha$, [Assumption Condition 2](#) yields

$$|L(x_h, a_h, \zeta_h) - L(\tilde{x}_h, a_h, \tilde{\zeta}_h)| \leq \left(C_z + \frac{C_x}{\alpha}\right) \chi_\alpha^h \varepsilon. \quad (49)$$

Multiplying by γ^h and summing from $h = 0$ to $H - 1$ gives (44). The [finite-infinite](#) geometric series gives the [displayed closed form uniform bound](#) when $\gamma\chi_\alpha < 1$. $\square$

[Assumption Condition 3](#). The [transition, stress, stage risk](#), and [PC rollout action map continuous service transition and stage risk](#) are Lipschitz in the action argument on the compact feasible set. [There are The PC allocation map \$\pi_c\$ before execution is Lipschitz in the physical state and threat state. There are finite constants \$L_a\$, \$\bar{L}_a^q\$, \$C_a\$, \$L_\pi^x\$, and \$L_\pi^z\$ such that](#)

$$\|F_c(x, a, \zeta) - F_c(x, \tilde{a}, \zeta)\| \leq L_a \|a - \tilde{a}\|, \quad (50)$$

$$|L(x, a, \zeta) - L(x, \tilde{a}, \zeta)| \leq C_a \|a - \tilde{a}\|, \quad (51)$$

$$\|\pi_c(x, \zeta, A) - \pi_c(\tilde{x}, \tilde{\zeta}, A)\| \leq L_\pi^x \|x - \tilde{x}\| + L_\pi^z \|\zeta - \tilde{\zeta}\|. \quad (52)$$

[The last inequality holds for PC rollout because the allocator in \(30\) has \$\epsilon > 0\$ and all score maps are continuous compositions of Lemma 5. Condition 3 holds for the continuous PC map on the stated compact domain.](#)

[Proof. Lemma 2 already establishes the first two inequalities in the action argument. It remains to prove the bound for \$\pi_c\$. For a score vector \$v \in \mathbb{R}_+^N\$, set \$r_i\(v\) = \sqrt{v_i} + \epsilon\$, \$S\(v\) = \sum_i r_i\(v\)\$, and \$p_i\(v\) = r_i\(v\)/S\(v\)\$. The mean value theorem gives](#)

$$\|r(v) - r(w)\|_2 \leq \frac{1}{2\sqrt{\epsilon}} \|v - w\|_2. \quad (53)$$

[Moreover, \$S\(v\) \geq N\sqrt{\epsilon}\$ and \$|S\(v\) - S\(w\)| \leq \sqrt{N} \|r\(v\) - r\(w\)\|_2\$. Since \$\|r\(w\)\|_2 \leq S\(w\)\$, let \$\Delta_S = |S\(v\) - S\(w\)|\$. Adding and subtracting \$r\(w\)/S\(v\)\$ gives](#)

$$\begin{aligned} \|p(v) - p(w)\|_2 &\leq \frac{\|r(v) - r(w)\|_2}{S(v)} + \frac{\|r(w)\|_2 \Delta_S}{S(v)S(w)} \\ &\leq \left(\frac{1}{2N\epsilon} + \frac{1}{2\sqrt{N\epsilon}}\right) \|v - w\|_2. \end{aligned} \quad (54)$$

[Equation \(30\) is an affine function of \$p\(v\)\$. Its Lipschitz constant is therefore at most](#)

$$(1 - \eta_B)B \left(\frac{1}{2N\epsilon} + \frac{1}{2\sqrt{N\epsilon}}\right). \quad (55)$$

[The fixed graph recurrent forecaster is a finite composition of affine maps, \[bounded projections\]\(#\), and \[square roots on a compact domain\]\(#\), the logistic function, and the hyperbolic tangent. Their Lipschitz constants are finite because the trained weights and \$A\$ are fixed. For every candidate, \(23\) is a finite composition of an affine map and a nonexpansive projection. Hence the finite horizon score \(26\) is Lipschitz in \$\(x, \zeta\)\$. Positive part preserves this property. The minimum of a finite collection of Lipschitz functions is Lipschitz with the largest of their](#)

[constants. The score vectors for charging, communication, central PC, and the robust extension are therefore Lipschitz. Combining their constants with \(54\) gives finite \$L_\pi^x\$ and \$L_\pi^z\$. This proof ends at the continuous allocation. Finite route selection, packet events, the OpenDSS switch, and integer crew completion are separate operations. \$\square\$](#)

[Corollary 4.1 \(continuous sensitivity before execution\). Let \$J_H^\pi\(x_0, \zeta_0\)\$ and \$\tilde{J}_H^\pi\(x_0, \tilde{\zeta}_0\)\$ be the closed-loop \$J_H^{\pi_c}\(x_0, \zeta_0\)\$ and \$\tilde{J}_H^{\pi_c}\(x_0, \tilde{\zeta}_0\)\$ be the predictive risks generated by PC rollout the continuous PC score and allocation layer from the same physical state and two initial threat estimates. Define](#)

$$\bar{L}_x = L_x + L_a L_\pi^x, \quad \bar{L}_z = L_z + L_a L_\pi^z, \quad (56)$$

$$\bar{C}_x = C_x + C_a L_\pi^x, \quad \bar{C}_z = C_z + C_a L_\pi^z. \quad (57)$$

For any $\alpha > 0$, let

$$\bar{\chi}_\alpha = \max \left\{ \|M\| + \|R\| L_\pi^z + \alpha \bar{L}_z, \bar{L}_x + \frac{\|R\| L_\pi^x}{\alpha} \right\}. \quad (58)$$

If $\|\zeta_0 - \tilde{\zeta}_0\| \leq \varepsilon$, then

$$|J_H^{\pi_c}(x_0, \zeta_0) - \tilde{J}_H^{\pi_c}(x_0, \tilde{\zeta}_0)| \leq \varepsilon \left(\bar{C}_z + \frac{\bar{C}_x}{\alpha} \right) \sum_{h=0}^{H-1} (\gamma \bar{\chi}_\alpha)^h. \quad (59)$$

[Proof. Let \$a_h = \pi\(x_h, \zeta_h, A\)\$ and \$\tilde{a}_h = \pi\(\tilde{x}_h, \tilde{\zeta}_h, A\)\$. \[Assumption \\$a_h = \pi_c\\(x_h, \zeta_h, A\\)\\$ and \\$\tilde{a}_h = \pi_c\\(\tilde{x}_h, \tilde{\zeta}_h, A\\)\\$. Condition 3 gives\]\(#\)](#)

$$\|a_h - \tilde{a}_h\| \leq L_\pi^x E_h + L_\pi^z \Delta_h.$$

[Combining this bound with \[Assumptions Conditions 1 and 3 yields and the restoration term in \\(23\\) yields\]\(#\)](#)

$$E_{h+1} \leq \bar{L}_x E_h + \bar{L}_z \Delta_h, \quad (60)$$

$$\Delta_{h+1} \leq (\|M\| + \|R\| L_\pi^z) \Delta_h + \|R\| L_\pi^x E_h. \quad (61)$$

For $V_h = \Delta_h + \alpha E_h$, the preceding two inequalities give $V_{h+1} \leq \bar{\chi}_\alpha V_h$. Since $V_0 \leq \varepsilon$, induction gives $V_h \leq \bar{\chi}_\alpha^h \varepsilon$. The stage risk difference satisfies

$$|L(x_h, a_h, \zeta_h) - L(\tilde{x}_h, \tilde{a}_h, \tilde{\zeta}_h)| \leq \bar{C}_x E_h + \bar{C}_z \Delta_h.$$

Using $E_h \leq V_h/\alpha$ and $\Delta_h \leq V_h$, then summing the discounted stage differences, gives the stated bound. $\square$

[Theorem 2 \(conditional finite portfolio regret\). Let \$\mathcal{P}\$ be the finite portfolio of integer crew routes available at one decision time. Let \$Q\(P\)\$ be the realized integrated risk of route \$P\$ and \$\hat{Q}\(P\)\$ its information feasible PC estimate. Suppose \$|Q\(P\) - \hat{Q}\(P\)| \leq \varepsilon_Q\$ for every \$P \in \mathcal{P}\$, and the route solver returns \$\hat{P}\$ satisfying](#)

$$\hat{Q}(\hat{P}) \leq \min_{P \in \mathcal{P}} \hat{Q}(P) + \delta_s. \quad (62)$$

[Then](#)

$$Q(\hat{P}) \leq \min_{P \in \mathcal{P}} Q(P) + 2\varepsilon_Q + \delta_s. \quad (63)$$

[The bound in \(63\) separates score approximation from the finite route solver tolerance. If the realized optimum \$P^*\$ is unique and](#)

$$\Delta_Q := \min_{P \in \mathcal{P} \setminus \{P^*\}} \{Q(P) - Q(P^*)\} > 2\varepsilon_Q + \delta_s, \quad (64)$$

then $\hat{P} = P^*$.

Proof. Let $P^* = \arg \min_P Q(P)$. Uniform approximation and solver tolerance give $Q(\hat{P}) \leq \bar{Q}(\hat{P}) + \varepsilon_Q \leq \bar{Q}(P^*) + \delta_s + \varepsilon_Q \leq Q(P^*) + 2\varepsilon_Q + \delta_s$. For any $P \neq P^*$, the gap condition implies $Q(P) > \bar{Q}(P^*) + \delta_s$, so no such plan can be returned. $\square$

Proposition 1 (execution feasibility and invariance). Assume the base feeder operating point is feasible, packet enactment satisfies $0 \leq d_{i,h} \leq 1$, service arrivals are bounded, $0 \leq \rho_q < 1$, and every accepted work order instance has a finite integer route. Station reassignment includes an unserved demand variable and enforces nonnegative station capacity. Then every implemented step satisfies the three resource budgets, packet reduction preserves those budgets, station flow respects capacity, the OpenDSS projection returns a feasible charging action, backlog remains bounded, and $\zeta_h \in \mathcal{Z}$ for all h .

Proof. Lemma 1 applies to each of the three allocators. Hence the requested actions are nonnegative and their component sums equal the corresponding budgets. For any component $u_i \geq 0$ and $d_i \in [0, 1]$, one has $0 \leq d_i u_i \leq u_i$. Summing this inequality proves that packet reduction cannot increase a budget. The feeder calculation selects a common scale $\alpha^{\text{AC}} \in [0, 1]$. The point $\alpha^{\text{AC}} = 0$ is the feasible base operating point. Bisection stores the feasible endpoint of every interval, so its returned endpoint is feasible and cannot increase the charging budget. In the station flow problem, every reassigned flow is constrained by the receiving capacity. Any remaining demand enters the nonnegative unserved variable. A feasible station allocation therefore exists and satisfies every capacity inequality.

Let $e_{i,h} \leq \bar{e}$. From the backlog recursion,

$$q_{i,h+1} \leq \bar{e} + \rho_q q_{i,h}, \quad (65)$$

$$q_{i,h} \leq \rho_q^h q_{i,0} + \bar{e} \frac{1 - \rho_q^h}{1 - \rho_q} \quad (66)$$

$$\leq \max \left\{ q_{i,0}, \frac{\bar{e}}{1 - \rho_q} \right\}. \quad (67)$$

The second line follows by induction, and the final line follows because it is a convex combination of $q_{i,0}$ and $\bar{e}/(1 - \rho_q)$. Thus backlog remains bounded. Suppose $\zeta_h \in \mathcal{Z}$. Projection maps $M\zeta_h + \xi_h$ into $\mathcal{Z}$. Every diagonal entry of $D(\bar{r}_h)$ is zero or one, so multiplication by $D(\bar{r}_h)$ also maps $\mathcal{Z}$ into $\mathcal{Z}$. Since $\zeta_0 \in \mathcal{Z}$, induction proves state invariance. Finally, a finite integer route assigns a finite arrival and service order to each accepted work order. The completion indicator changes only at the corresponding recorded completion time, which makes every realized repair update well defined. $\square$

Theorem 1 and Corollary 1 give a direct interpretation of the policy. The weighted growth factors χ_α and $\bar{\chi}_\alpha$ are determined by threat propagation, stress feedback, action sensitivity, and state sensitivity. Restoration is evaluated through the future threat sequence that enters the same finite horizon risk envelope. concern finite horizon sensitivity in the continuous prediction and allocation

layer. Theorem 2 gives a conditional bound for the finite route portfolio in terms of ε_Q and the solver tolerance. Proposition 1 covers constraint preservation in execution. The numerical checks in Section IV evaluate these same properties for the reported scenarios.

IV. Experimental Design and Results

IV. Experimental Study

A. Data construction

A. Evidence sources and preprocessing

The experiments use public yellow taxi trip records released by the New York City Taxi and Limousine Commission to construct a large mobility driven charging load data set. The complete year The evaluation separates measured demand, outage context, electrical feasibility, communication enactment, and crew execution instead of deriving all layers from one mobility proxy. Figure 1 shows the resulting closed loop, and Table I records the role of each source.

The primary demand source is the City of Boulder Electric Vehicle Charging Station Data under a CC0 license. Its metadata defines one row as one transaction and provides local transaction time, station identity, charging duration, and delivered energy. The local archive contains all 148,136 transactions released from 2018 through 2023 collection contains 38, with 50 station records and 1,310,252,226 raw trip records. Invalid location identifiers, trips outside the calendar year, and implausible trip distances are removed. This leaves 36, 879, 930 valid trips. Trips are aggregated to hourly values over 263 zones, which gives 8760 time steps. For each hour and zone, the data construction computes pickup activity, drop off activity, trip mileage, charging energy demand, and origin-destination flows 719,654 measured positive kWh. Four records lack an end time, 16,113 report zero energy, no duplicate source identifier or full row was found, and no negative energy was retained. Arrival counts are exact transaction counts by start hour. Session energy is measured. Its hourly load profile is obtained by spreading each session's energy uniformly over the reported charging duration. Conservation checks recover all 148,136 arrivals and the measured positive energy to within 0.001 kWh.

Let $d_{\tau i}^{\text{mile}}$ denote the total trip distance assigned to zone i in hour τ . Charging energy is estimated from this mileage by where $\pi_{\text{EV}} = 0.18$ is the assumed electric vehicle share and $\epsilon_{\text{mile}} = 0.24$ kWh per mile is the energy use coefficient for electric mobility. The factor π_{EV} converts observed travel mileage into electric mobility mileage, and ϵ_{mile} converts electric mobility mileage into charging energy. The same conversion is applied to all methods. The origin-destination matrix is converted to a top neighbor graph and normalized. Feeder capacity and communication capacity indicators are deterministic functions of historical load intensity. The resulting data set is large, public, and directly reproducible in the digital twin.

Data construction used by the digital twin Quantity Value Raw trips 38, 310, 226 Valid trips after filtering 36,

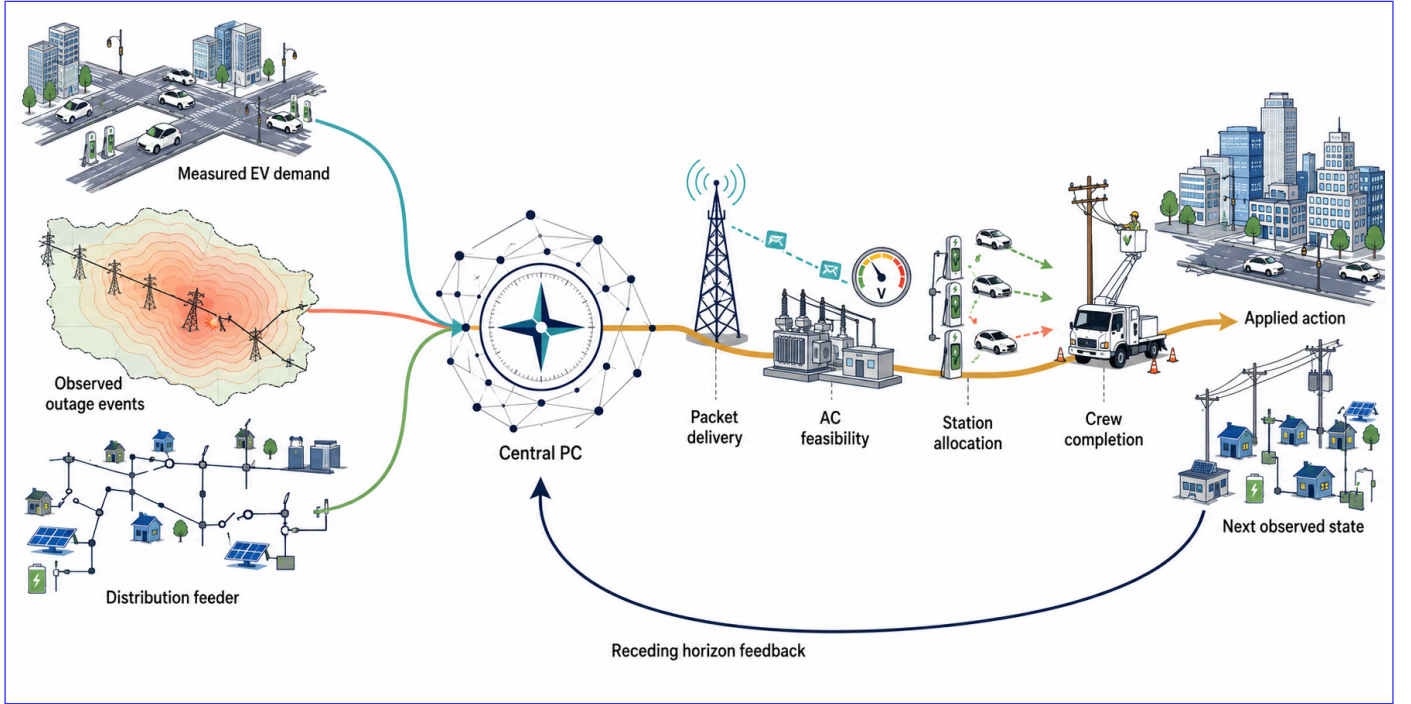


Fig. 1. Executable PC closed loop. Boulder charging sessions supply measured demand and forecasts. EAGLE-I supplies county level outage calibration. SMART-DS independently constrains feeder action. PC rollout issues requested actions, while service and threat states change only after distribution system packet deadline enactment, distribution system projection, station flow, and integer crew routing. The next observation restarts the hourly horizon. The three public sources serve distinct model roles.

TABLE I
Independent evidence sources and their distinct roles

Source	Scale	Observed fields	Role in this study
City of Boulder EV sessions	148,136 transactions from 50 stations between 2018 and 2023	Arrival time, charging duration, energy (kWh), station and address	Measured charging demand with calendar periods for training, selection, and testing
EAGLE-I	249,316,543 national rows with 819,528 retained in Boulder and six adjacent counties between 2014 and 2025	County, UTC timestamp, customers out	Power state persistence, event and recovery distributions, and uncertainty bounds with missing rows preserved
SMART-DS SFO P1U feeder	80 buses, 219 nodes, 27 loads, 67 lines, 13 transformers	OpenDSS circuit, equipment and peak load operating point	Online projection for all primary policy hours and the integrated electrical stress test
Packet and route models	12,000 packet stress rows and 4,096 paired main scenarios with up to 36 jobs and four crews	Queue events, retries, deadlines, backup, integer crews, travel, service and completion times	Requested controls and threat updates after a recorded repair completion event

879, 930 Zones 263 Hourly states 8760 Historical window 12 hours Rollout horizon 6 hours Electric vehicle share π_{EV} 0.18 Energy use coefficient c_{mile} 0.24 kWh per mile

Table I summarizes the quantities that enter the digital twin. The raw trip records are reduced to an hourly state tensor, but the tensor still preserves demand variation from a full year of city scale mobility records. The charging and communication quantities remain fixed deterministic transformations of the same processed activity data, which keeps the policy comparison tied to one common operating environment.

B. Compared policies

The forecasting history is twelve hours and the rollout horizon is six hours. The chronological data split uses the first part Time was split before model fitting. The period from 2018 through 2021 was used for training, the middle part for validation, and the final part for testing. The graph recurrent predictor is trained on a graphics processor. The tested operating policies are static allocation, greedy current state allocation, neural rollout, PC rollout, and an oracle reference. Static allocation uses historical averages. Greedy allocation reacts only to the current grid and charging state. Neural rollout uses the same forecast as PC rollout and assigns restoration from the current threat field. The oracle uses

future realized demand and threat values and is included only as a reference. 2022 was used for model selection, and 2023 was used for final testing. No random row split crosses this calendar boundary. A second five fold spatial test withheld contiguous west to east station blocks from target fitting. In that inductive deployment test, a held out station's calibration before 2022 and observed 1, 24, and 168 h lags were permitted, but its target values never entered coefficient fitting.

Threat scenarios are generated in four groups. The nominal group contains small perturbations. The single domain group perturbs one of mobility, power, or communication. The cascading group allows cross domain propagation. The unseen compound group uses stronger and longer disruptions than those used in the nominal and single domain cases. Each group contains the same number of scenarios. Every policy is evaluated on the same realized demand and threat sequence. Power event evidence came from the CC BY 4.0 EAGLE-I release [37]. The source record contained 17 files and 249,316,543 rows. Preprocessing retained 819,528 observations for Boulder County and its six adjacent counties, including 171,187 Boulder observations. EAGLE-I timestamps are UTC and were converted to America/Denver only for alignment with local charging records. Because the source omits zero outage records and cannot distinguish a true zero from a collection gap, absent county and time pairs were preserved as missing rather than silently imputed as zero. Events were evaluated at 50, 100, and 500 customers out. At the 50-customer threshold, 2,535 events had median duration 1.75 h and 90th percentile duration 6.5 h. The post peak half recovery time was observable for 19.6% of events and had median 0.75 h. These traces inform a scenario distribution, not station level failure labels or utility work orders.

The threat generator selects initial affected zones with probabilities proportional to historical demand. This makes disruptions more likely to enter busy charging service zones while still allowing all zones to appear across the scenario set. For scenario class g , the latent threat intensity follows where κ_0 is temporal persistence, ρ_g controls spatial spread, and ν_h is a bounded multiplicative perturbation. Mobility, power, and communication threat fields are obtained by domain specific scaling of b_h . In cascading and unseen compound scenarios, cross domain terms are added before projection. The coefficients τ_{pc} , τ_{rc} , τ_{cr} , and τ_{cp} specify power to communication, mobility to communication, communication to mobility, and communication to power propagation strength. independent electrical test uses one complete SMART-DS v1.0 San Francisco P1U feeder and its unmodified OpenDSS model. The solved circuit contains 80 buses, 219 nodes, 27 loads, 67 lines, and 13 transformers. Eighteen nodes already deenergized in the released base case are identified and excluded from active voltage minima. They are not counted as failures caused by EV action. The base active node voltage minimum is 0.989 p.u. and maximum line loading is 0.913 p.u. Requested EV

TABLE II
Calendar split test performance on measured Boulder charging sessions in 2023

Method	Arrivals		Energy (kWh)	
	MAE	RMSE	MAE	RMSE
Historical mean	0.137	0.517	1.101	3.026
Weekly seasonal	0.156	0.637	1.130	3.299
Persistence	0.175	0.678	1.059	3.186
Graph recurrent forecaster	0.110	0.527	0.945	3.113

actions are aggregated as balanced three phase, wye connected constant PQ increments at the 17 eligible three phase loads. Active power is the executed station request and reactive power follows a 0.98 lagging power factor. The scenario generator follows the propagation model in (23), while the controller receives the realized state and applies its own rollout rule. The shared scenario stream ensures that differences in Table V come from policy decisions rather than from different disturbance draws. released load voltage level (0.48 or 12.47 kV) and base load parameters are otherwise unchanged. Twenty station to load mappings are prepared in advance. Mapping $s \bmod 20$ is shared by all six policies in scenario s and recorded in the scenario table. At every simulated hour, station level executable charging is aggregated at the assigned loads. OpenDSS solves the unprojected action, and a bisection backtracks its common fraction until voltage lies in $[0.95, 1.05]$ p.u., line loading is at most 1.0 p.u., and the circuit converges. Only this projected energy enters served energy and backlog equations. This gate validates the electrical feasibility of a proposed charging action. It does not simulate feeder topology reconfiguration, switching or breaker operations, fault isolation, feeder resupply, or equipment parameter changes after repair. The experiment therefore studies charging service restoration under a fixed feeder topology.

For each scenario, all five policies are evaluated on the same demand horizon and threat horizon. The reported cost is the sum of mobility delay, unserved charging demand, communication loss, feeder margin violation, and cascade interaction terms appearing in (14). Service continuity is computed as served mobility divided by total mobility over the horizon. The final tables report mean values across all evaluated scenarios, and the scenario class table keeps the same aggregation within each threat group.

B. Demand forecasting and spatial transfer

Policies compared in the rollout evaluation Policy Decision information Static Historical average service and restoration allocation Greedy Current feeder, charging demand, and current threat field Neural rollout Graph forecast with current threat restoration PC rollout Graph forecast with future cascade envelope restoration Oracle Realized future demand and threat values for reference

Table II reports the fixed 2023 temporal test. The graph recurrent forecaster achieved arrival MAE 0.110 and

TABLE III
Propagation coefficients and prespecified uncertainty set

Parameter	Central	Interval	Evidence class
ρ_r	0.728	[0.546, 0.911]	estimated
σ	0.233	[0.116, 0.349]	estimated
ρ_p	0.772	[0.754, 0.787]	estimated
τ_{pr}	0.000	[0.000, 0.032]	estimated observational
τ_{pc}	0.166	[0.012, 0.271]	constructed
ρ_c	0.500	[0.300, 0.800]	sensitivity only
τ_{rc}	0.040	[0.000, 0.080]	sensitivity only
τ_{cr}	0.080	[0.000, 0.160]	sensitivity only
τ_{cp}	0.040	[0.000, 0.080]	sensitivity only

energy MAE 0.945 kWh, compared with 0.137 and 1.101 kWh for the historical mean. Its energy RMSE (3.113 kWh) did not beat the historical mean (3.026 kWh), so no uniform metric dominance is claimed. Across three training runs, arrival MAE ranged from 0.1096 to 0.1099 and energy MAE from 0.9452 to 0.9468 kWh.

The spatial test pooled all five held out blocks and six horizons, the spatial ridge model reduced energy MAE from 1.130 to 1.054 kWh and RMSE from 2.928 to 2.272 kWh relative to station specific weekly history. Arrival RMSE also decreased from 0.557 to 0.440, but arrival MAE (0.161) was worse than the pooled calendar mean (0.139). Thus the measured data forecast supports temporal deployment and improves several unseen station endpoints. Arrival MAE remains higher at sparse stations, which identifies the forecast condition for interpreting the decision study.

C. Propagation calibration and policy comparison

The coefficient ledger in Table II clarifies the main comparison. Neural rollout and PC rollout have the same forecast input. Their difference is the restoration allocation. This design keeps the experiment aligned with the analysis in Theorem 1 and Corollary III distinguishes estimated, observational, constructed, and sensitivity range quantities. A nonnegative station deficit regression estimated road and mobility persistence and spatial spread from 1, where the performance gap is connected to future threat growth rather than to a different demand estimator. 268,661 training samples. On 3,108 aligned 2023 test hours, its road state RMSE was 0.306 versus 0.374 for persistence. EAGLE-I estimated power persistence at 0.772 with a day block bootstrap interval [0.754, 0.787]. The aligned coefficient from power state to EV deficit was effectively zero with upper interval 0.032, providing no evidence for a strong direct conditional association. The packet model constructed the power to communication interval. Directions without joint field traces span sign constrained sensitivity ranges including zero.

D. Computation

The experiments require two types of computation. The graph forecasting model uses parallel tensor operations on a graphics processor. The scenario evaluation uses many

independent rollout simulations and is therefore executed with parallel central processor cores. This division keeps the graphics processor busy only during model training and moves scenario evaluation to central processor cores. The final evaluation includes thousands of base scenarios and an additional set of power oriented stress cases. Reduced examples can be run on ordinary hardware, but the complete repeated experiment is substantially more convenient on a shared computing platform with both graphics and central processor resources. The primary central PC policy evaluates multistep propagation under the calibrated central matrix. The robust extension evaluates 21 central/endpoint matrices and allocates using the smallest predicted marginal action benefit. In the realized dynamics, each coefficient was sampled independently from its prespecified interval for each scenario. Forecast matched, central PC, and robust PC received the same selected demand forecast, current threat observation, backlog, and realized past. Static instead used historical station means, greedy used the common forecast but formed a direct current threat route, and the oracle alone received future innovations. The forecast matched baseline uses no propagation matrix. It scores the common finite route portfolio from current threat and forecast exposure only. Central PC propagates the calibrated central matrix, whereas robust PC minimizes the worst score over the 21 matrix set. Thus forecast matched is not central PC. Table IV fixes the information and execution contract for all six policies. The oracle alone receives future innovations. The primary family comprised central PC versus the matched baseline for total cost overall and in four prespecified threat classes. We used 4,096 paired scenarios (1,024 per class), paired bootstrap intervals with 20,000 resamples, paired t tests, two sided Wilcoxon tests, and Holm correction across the five classwise tests. Supplementary Table S1 reports each value, unit, role, calibration rule, selection source, and tested range.

Training and rollout computation settings Quantity Value Trainable parameters 30

Table V reports the primary paired comparison and the policy summary. Central PC reduced mean total cost by 0.6327% overall relative to the matched baseline (paired difference -6.2730, 220 Training epochs 80 Training repetitions 3 Scenario classes 4 Scenarios per class 1024 Policies per scenario 5 Power stress and sensitivity cases 9 Central processor cores for evaluation 32 Base evaluation simulation time 1.10 seconds 95% bootstrap CI [-7.0372, -5.5195]). The Holm adjusted paired t and Wilcoxon probabilities were 3.84×10^{-56} and 9.76×10^{-101} , respectively. All four prespecified groups had negative intervals, with reductions of 0.0020% in nominal, 0.0433% in single domain, 0.6502% in cascade, and 1.1296% in OOD compound cases. The nominal effect is statistically distinguishable in these simulated pairs but operationally negligible. The intervals quantify Monte Carlo scenario uncertainty, not variation across real utility systems.

TABLE IV

Policy information and execution contract. “Current” means the observation available at the decision hour. The common route portfolio contains routes induced by static, greedy, forecast matched, central PC, robust PC, and uniform priorities. Future innovations are never revealed to deployable policies.

Policy	Demand information	Threat and backlog	Multistep transition rollout	Transition matrix used	Future innovations	Repair route rule	Common execution gates
Static	Historical station mean	Backlog only for charging	No	None	No	Direct route from static priority	Yes
Greedy	Common forecast	Current threat and backlog	No	None	No	Direct route from current threat priority	Yes
Forecast matched	Common forecast	Current threat and backlog	No	None	No	Current exposure score over common portfolio	Yes
Central PC	Common forecast	Current threat and backlog	Yes	Calibrated central M	No	Central transition score over common portfolio	Yes
Robust (extension)	PC Common forecast	Current threat and backlog	Yes	Worst score over 21 declared matrices	No	Worst transition score over common portfolio	Yes
Oracle	Realized future demand	Current threat and backlog	Yes	Central M	Yes	Direct route from oracle priority	Yes

TABLE V

Primary transition comparison and policy cost summary under sampled uncertainty. Positive reduction denotes lower cost than forecast matched. C/M gives the central PC and matched mean costs. The oracle alone receives future demand and innovations.

Threat group	Pairs	C/M cost	Reduction (%)	Δ 95% CI
All	4096	985.17/991.44	0.6327	[-7.0372, -5.5195]
Nominal	1024	661.82/661.83	0.0020	[-0.0233, -0.0053]
Single domain	1024	689.06/689.35	0.0433	[-0.4778, -0.1179]
Cascade	1024	985.30/991.75	0.6502	[-7.5902, -5.3153]
OOD compound	1024	1604.49/1622.82	1.1296	[-21.0397, -15.6852]

Policy mean over the same 4,096 scenarios		
Policy	Mean cost	Reduction (%)
Static	1069.50	-7.873
Greedy	992.07	-0.063
Forecast matched	991.44	0.000
Central PC	985.17	0.633
Robust PC	986.85	0.463
Oracle	896.01	9.626

The computation settings in Table III explain why the evaluation was run as a repeated parallel experiment rather than as a small example. The forecasting model is modest in size, while the evaluation requires repeated closed loop rollouts over many disruptions. The runtime is controlled by the independence of scenario rollouts and by distributing them over central processor cores. Robust extension had higher mean cost than central PC (986.85 versus 985.17). The robust minus central difference was 1.6830 with 95% CI [1.3298, 2.0508], or a 0.1708% penalty. Its 95th percentile cost was equal to central PC (2381.08), its 99th percentile was 7.47 lower, but its CVaR at 95% and 99% was 4.48 and 3.19 higher, respectively, and the maximum was equal (4084.41). Robust PC won 3.83%, tied 88.50%, and lost 7.67% of paired scenarios. These diagnostics do not support either an average cost or a coherent tail risk advantage for worst set scoring, so robust PC is retained only as a sensitivity extension. Central PC reduced cost by 7.885% relative to static allocation. The oracle alone received future innovations. Its mean cost was 896.01, leaving central PC with a 9.95% policy to oracle gap. The oracle is therefore an information value reference

TABLE VI

Objective sensitivity of central PC relative to forecast matched on the same 4,096 paired trajectories. Policies retain their original actions under alternative weights.

Evaluation weights	Reduction (%)	Paired difference (95% CI)
Reference weights	0.6327	-6.2730 [-7.0372, -5.5195]
2× mobility weight	0.6375	-6.3821 [-7.1811, -5.6118]
2× unserved energy weight	0.4105	-6.9247 [-7.7991, -6.0867]
2× communication weight	0.7978	-8.9713 [-10.0243, -7.9314]
2× power service weight	0.6411	-6.5376 [-7.4521, -5.6625]
0.5× cascade weight	0.5380	-4.9984 [-5.6519, -4.3621]

rather than evidence of near optimality.

D. Empirical results

Table VI shows that changing one objective weight at a time did not reverse the central PC comparison with forecast matched. The reduction ranged from 0.4105% to 0.7978%, and every paired bootstrap interval remained below zero. These tests on fixed trajectories assess evaluation weight dependence. They do not claim that a policy optimized again for each alternative objective would select the same route.

Figure 2 places the primary transition comparison beside the electrical, communication, and crew execution results. The panels use the same values and sample definitions as the corresponding tables.

Mean cost and service interruption for the compared policies.

D. Electrical and mobility execution

With the forecast fixed, Table VII summarizes the feeder and crew outcomes in the main loop. The primary run with 4,096 scenarios made 147,456 online OpenDSS calls, one for each scenario, policy, and hour. At measured load, all raw actions were already feasible, so projection retained 100% of requested charging. This feasibility result but cannot by itself demonstrate feedback. The companion 20× stress run therefore used 1,024 paired scenarios and the same loop. It produced 1,089 infeasible policy hours.

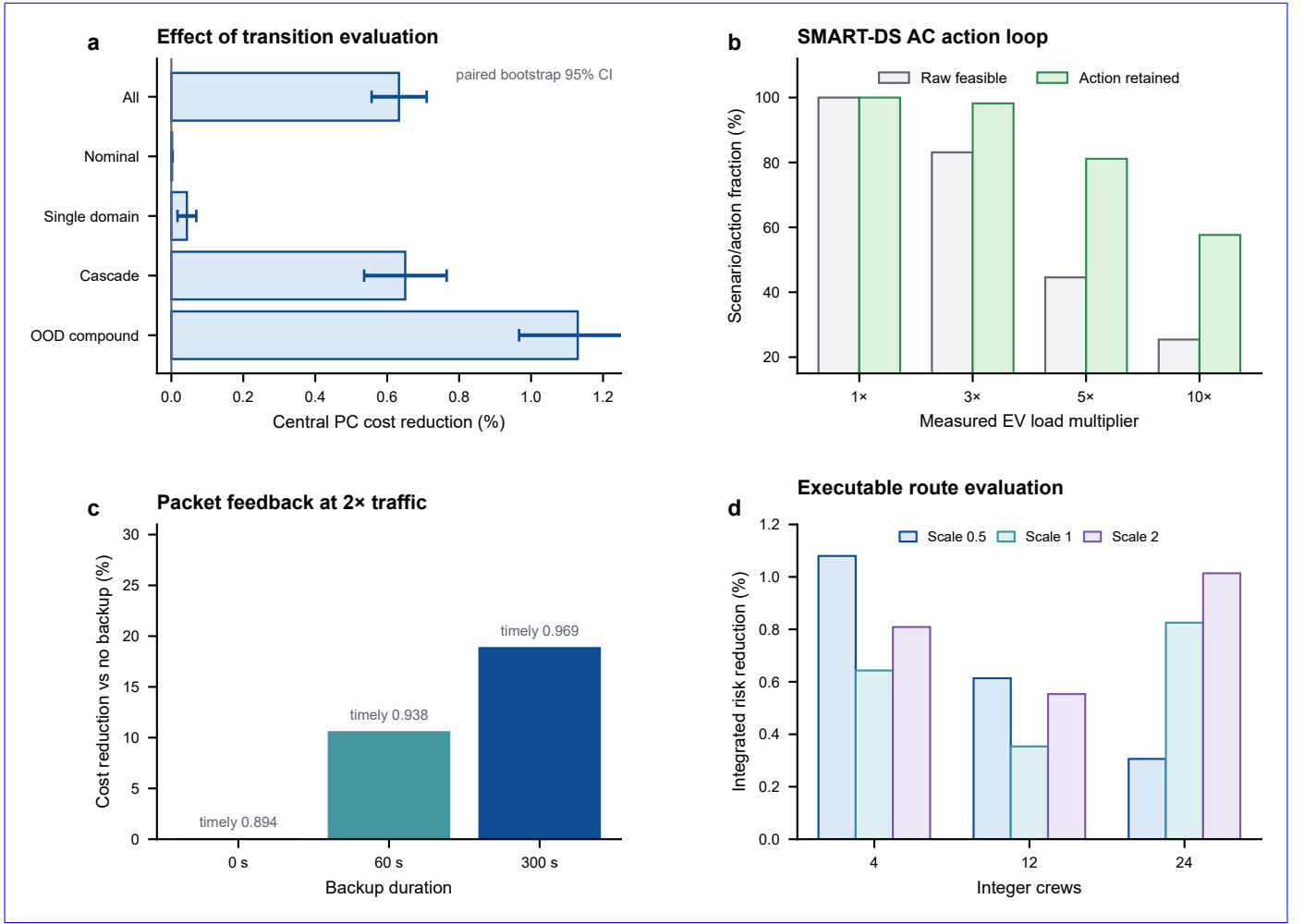


Fig. 2. Decision and execution evidence. Panel (a) reports the central PC cost reduction relative to the forecast matched baseline. Filled intervals show paired means and bootstrap confidence limits for each threat class and for all scenarios. Panel (b) reports raw distribution system feasibility and the fraction of requested EV action retained after projection. Panel (c) reports the cost reduction from backup power under increased packet traffic, with the mean fraction of actions delivered on time printed above each bar. Panel (d) reports integrated risk reduction when executable routes are evaluated before selection. Exact sample sizes, confidence intervals, and adjusted tests are reported in the surrounding tables.

TABLE VII
SMART-DS projection and routed crew events inside the closed loop evaluation

Condition	Pairs	AC hours	Raw infeas.	After proj.
Measured load	4,096	147,456	0	0
20x EV stress	1,024	36,864	1,089	0

Condition	Action retained (%)	Crew completed (%)
Measured load	100.00	81.86
20x EV stress	99.26	81.92

Projection reduced this count to zero, retained 99.26% of requested service on average, and placed 69,664.8 kWh of curtailed energy into unserved demand/backlog rather than crediting it as served. Thus the online adapter changes downstream state whenever the AC limit binds.

Table VIII extends the electrical comparison across load multipliers. The SMART-DS experiment comprised 20

TABLE VIII
SMART-DS AC action projection (20 mappings, 24 hours each)

EV load	Raw feasible (%)	Action retained (%)	Raw $V_{\min}$	Raw line p.u.
1x	100.0	100.0	0.986	0.929
3x	83.1	98.2	0.977	0.968
5x	44.6	81.1	0.966	1.044
10x	25.4	57.7	0.938	1.361

randomized station to load mappings, 24 test hours, and four EV load multipliers (1,920 cases). Raw actions were all feasible at measured load. At 3x, 5x, and 10x load, raw feasibility fell to 83.1%, 44.6%, and 25.4%, whereas the action projection retained 98.2%, 81.1%, and 57.7% of requested EV power. Every projected case converged and met the voltage and line limits. A separate test for all of 2023 used all 7,451 nonzero hours at 1x and 3x load (14,902 cases). At 3x, 99.17% of raw actions were feasible and projection retained 99.94% on average. All projected cases remained feasible. The reported curtailment is

TABLE IX
Station choice under measured demand for 24 peak hours

Closure (%)	No choice	Nearest	Logit	Coordinated	Extra km
10	67.4	100.0	100.0	100.0	0.32
20	55.8	100.0	100.0	100.0	0.50
30	43.7	100.0	100.0	100.0	0.97

TABLE X
Backup power effect with packet feedback at $2\times$ traffic with
 $n = 1024$ paired scenarios

Backup	Timely action	Restoration enacted	Cost reduction (%)	Difference 95% CI
60 s	0.938	0.756	10.54	[-147.3, -126.8]
300 s	0.969	0.789	18.83	[-261.2, -227.7]

therefore produced inside the action path and not inferred from a later voltage label.

Table IX reports the station reassignment results. Station closure does not automatically imply abandoned charging. A separate capacity constrained flow model distinguished local service, rerouting, and unserved demand across 24 observed peak hours. Under measured demand and 10, 20, and 30% closures, preserving the original station served only 67.4, 55.8, and 43.7% of energy. Nearest, logit, and coordinated reassignment all served 100% in these cases. The coordinated solution required only 0.32, 0.50, and 0.97 km energy weighted extra distance. Under $4\times$ demand, total constructed capacity, rather than the choice model, became the limiting factor. This stress result prevents the rerouting mechanism from creating fictitious capacity.

E. Communication and crew execution

The packet emulator is a finite buffer queue with first come, first served discipline, discrete events, erasures, two retransmissions, backoff, a 500 ms control deadline, and explicit backup duration. An M/M/1 unit test at 10 packet/s arrival and 20 packet/s service reproduced the theoretical 100 ms mean latency with 0.269% relative error. Across 12,000 station stress cases, the original algebraic support proxy had Spearman correlation 0.860 but mean absolute error 0.243 against the simulated fraction of actions delivered on time, which confirms that the proxy cannot replace packet dynamics.

Table X reports the effect of backup duration. Equation (8) closes this evidence loop. Under prespecified $2\times$ traffic and 1,024 paired uncertain transition scenarios, 60 s backup raised the mean timely action fraction to 0.938 and reduced central PC cost by 10.54% relative to no backup (difference CI [-147.3, -126.8]). A 300 s backup raised the fraction to 0.969 and reduced cost by 18.83% (difference CI [-261.2, -227.7]). Before backup exhaustion, power threat caused no direct service rate derating. The effect arose only after the coded depletion event.

In the integrated primary run, the route generated for four crews was not a diagnostic added after scoring. The same work order set was dispatched through packet

TABLE XI
Route evaluated PC compared with matched routing

Crews	Service scale	Risk reduction (%)	Difference 95% CI	Holm p
4	0.5	1.08	[-2.25, -0.47]	0.0237
4	1.0	0.64	[-2.33, 0.06]	0.177
4	2.0	0.81	[-2.85, -0.94]	0.00175
12	0.5	0.61	[-0.65, -0.12]	0.0492
12	1.0	0.35	[-0.68, 0.00]	0.129
12	2.0	0.55	[-1.63, -0.20]	0.00652
24	0.5	0.31	[-0.30, 0.01]	0.0293
24	1.0	0.83	[-0.94, -0.37]	6.49e-05
24	2.0	1.01	[-1.95, -0.82]	6.91e-06

delivery, and threat was updated only at its recorded completion hours. Central PC completed 82.17% of dispatched jobs versus 81.72% for the forecast matched baseline, reduced mean job completion time from 3.131 to 3.095 h, and reduced crew travel from 2.975 to 2.957 h per scenario. Paired bootstrap intervals for the differences were [0.0038, 0.0051], [-0.0417, -0.0300] h, and [-0.0263, -0.0102] h, respectively.

Table XI reports the route comparison for every crew and service condition. The route experiment used 36 jobs, integer fleets of 4, 12, and 24 crews, geocoded travel at 30 km/h with 1.25 circuitry, and service time multipliers 0.5, 1, and 2. Each repair completion causally removed that job's risk from the differences in Table V and Fig. 2 come mainly from restoration decisions. Service continuity is not the most discriminating measure in the aggregate view because many scenarios are nominal or single domain, accumulated state. The original PC priority lost its advantage in some integer routed cases. Route evaluated PC corrected this result by selecting the lowest predicted risk executable candidate rather than projecting a continuous priority after the decision. In 128 independent scenarios per cell, route evaluated PC reduced mean integrated risk in all nine crew and service conditions by 0.31% through 1.08%. Six bootstrap intervals excluded zero and seven Holm adjusted Wilcoxon tests were below 0.05. The intervals for four crews at scale 1.0, twelve crews at scale 1.0, and most policies can still keep a large share of mobility service available. Total cost separates the methods more clearly because it also counts charging backlog, communication loss, feeder margin violation, and cascade interaction. Static allocation is stable but spends resources in zones with little current or future exposure. Greedy allocation reacts to present stress without modeling threat entering from neighboring zones. Neural rollout uses future demand for service placement, and PC rollout keeps that demand information while changing the restoration side of the decision. The lower cost is therefore tied to the future cascade envelope in (26), not to a stronger predictortwenty four crews at scale 0.5 included zero and remain visible in Table XI. The observed gain is not declared uniformly significant. The Holm adjusted Wilcoxon value for twenty four crews at scale 0.5 is below 0.05 while its bootstrap interval crosses zero because the two procedures test different statistics. The interval targets the paired mean

difference, whereas Wilcoxon ranks the nonzero paired differences and is affected differently by ties and skewness. We therefore require the mean interval, not either test in isolation, for a mean improvement claim.

The class split in Table VI gives the mechanism behind the average result. Nominal cases leave little room for a future propagation model to help, and single domain cases are largely handled by current state reaction. The gap widens in cascading and unseen compound cases, where a zone may have moderate current loss but high future exposure because it lies between a power disturbance and a communication sensitive mobility corridor. This operating condition is exactly the case targeted by the restoration score. A demand only rollout and a current state greedy rule can keep serving present load, yet both can miss part of the value of repairing a zone before it becomes a cross domain transmission point. integrated loop sensitivity runs used another 1,024 paired scenarios per condition. Central PC reduced cost relative to matched by 0.529%, 0.421%, and 0.668% under 0.75M, 1.25M, and 0.15 coefficient noise, respectively. The robust minus central difference was -0.470 at 0.75M with 95% CI $[-1.034, 0.028]$, but $+6.383$ at 1.25M with CI $[4.910, 7.976]$ and $+2.272$ under coefficient noise (CI $[1.547, 3.069]$). Thus the only favorable overall robust comparison was not statistically distinguishable from zero, whereas two misspecification settings favored central PC. With 12 crews robust PC was 0.075 cost units worse than central PC with CI $[0.005, 0.165]$. With 24 crews its 0.022 cost unit advantage was negligible and the interval crossed zero.

F. Numerical and mathematical checks

Feeder margin violation and unserved charging demand under grid stress cases. The feeder margin panel uses $\log_{10}(1 + \text{value})$. Eleven independent checks connected the mathematical statements to the numerical calculations. Ten thousand random score and budget tests preserved each continuous budget, with a maximum numerical error of 1.37×10^{-4} . Another 5,000 largest remainder tests preserved the integer crew count. All 8,117 nonzero 2023 test hours satisfied the analytical zero service backlog bound. A further 100,000 steps at simultaneous upper uncertainty bounds remained inside $[0, 0.98]^{3N}$. All 14,902 standalone SMART-DS projections, 147,456 primary online calls, and 36,864 stress online calls were feasible after projection. Packet backup and action derating checks had zero logic error. Every checked route visited each selected job once. All main crew counts matched completion event logs with paired inputs. All 2,592 station choice flows met capacity. The supporting map links each property to its theorem, function, evidence file, and status.

The grid stress cases extend the comparison beyond average cascade behavior. In Table VII and Fig. 3, charging demand is increased and feeder margin is reduced, so feeder violation and unserved charging both rise with stress level. PC rollout remains below neural

rollout in cost across these cases, although the relative gain becomes smaller under the most severe feeder stress. That change is important for interpreting the result. Restoration can reduce propagation through the communication and mobility layers, but it cannot create feeder capacity removed by the stress setting. The coupling cases give the complementary view. Stronger power to communication and communication to mobility coupling raises total cost, while the PC rollout gain remains present because the envelope still shifts restoration toward zones likely to transmit disturbance across domains.

V. Interpretation and Reproducibility

Changing the restoration budget

A. Result interpretation

The paired comparison identifies the value of transition evaluation rather than the value of an additional demand forecast. Central PC and the forecast matched baseline receive the same forecast, current state, feeder mapping, work orders, route candidates, and realized history. Their difference is that central PC propagates the calibrated transition before selecting a route. The overall improvement is modest, becomes larger in cascade and compound cases, and remains in the same direction under the tested objective weights. The small nominal effect is consistent with the decision rule because future propagation changes little when the initial threat is mild.

The robust extension provides a separate comparison with central PC. It has a higher mean cost and does not reduce the reported conditional tail means or the maximum cost. The direct comparisons under altered transition coefficients also favor central PC in the settings with resolved intervals. These results support the calibrated central transition as the principal model and retain the finite worst set calculation as a sensitivity variant. The distinction also keeps the empirical conclusion aligned with the policy definitions in Table VIII adds a second check on the same mechanism. A larger budget lowers cost and improves service continuity, while feeder violation changes only moderately because local capacity and exogenous charging demand still constrain that term. The largest benefit comes from reducing communication loss and mobility delay during cascading disturbances. This ordering matches Theorem 1 and Corollary 1, which identify propagation growth, stress feedback, and action sensitivity as the quantities that control finite horizon risk. PC rollout does not alter the demand forecast used by neural rollout. It changes restoration so that future cross domain exposure is smaller before the cascade becomes visible in the current loss map. IV.

Execution results explain the route from a requested decision to a realized state. At measured load, the feeder accepts every raw charging action. Under the electrical stress condition, projection reduces the request and transfers curtailed energy to backlog. The

packet experiment shows that backup duration changes the fraction of commands delivered on time and therefore changes operating cost. The crew experiment records repair only after an integer route reaches the corresponding completion time. Electrical feasibility is thus calculated independently after policy scoring, while relative policy cost is estimated from the stated service objective and transition scenarios.

The three public data sources enter through distinct variables. Boulder transactions provide measured station arrivals and session energy. EAGLE-I provides county level event timing for transition calibration and service time scenarios. SMART-DS provides an independent feeder for charging feasibility. Communication traffic and backup duration define controlled operating conditions. This arrangement forms a public data service restoration study with a fixed feeder topology and explicit execution rules.

B. Data and code availability

A public repository provides the code, processed inputs, scenario level outputs, trained model parameters, statistical results, distribution feeder, route completion records, and packet records used in this study. Source metadata record the public origin, license, access date, byte count, and SHA256 value of each reused data collection. Processed data retain variable definitions, calendar splits, missing value rules, time zone conversion, and energy conservation checks. The repository also includes the software environment, fixed initialization records, complete parameter table, and the evidence needed to regenerate the principal statistics. The repository is available at <https://github.com/ahshicr/TSG-01697-2026-reproducibility>.

VI. Conclusion

We studied rollout restoration for cyber-physical smart grids coupled with urban electric mobility. PC rollout combines graph-based charging state forecasting, a physics guided digital twin, a monotone threat propagation. This paper connects measured charging demand, a calibrated transition model, and a concave restoration allocation rule. The theoretical analysis gives a finite horizon risk bound that connects rollout performance to the propagation growth factor. The public city scale mobility experiment indicates that the method is most useful in cascading, compound, and stressed charging operation, while normal operation remains close to the neural rollout baseline. Full year training and repeated closed loop scenario evaluation benefit from parallel graphics and central processor resources executable restoration decisions. Central PC evaluates the future effect of a candidate route while a matched baseline uses the same forecast and current state without threat propagation. The paired results show a consistent cost reduction for central PC, with the clearest effect under cascade and compound conditions. The worst set extension does not improve the reported mean or tail measures and is therefore retained as a sensitivity

variant. Packet delivery, distribution system projection, station capacity, and integer crew completion determine whether a request changes the realized state. The mathematical results establish finite horizon sensitivity for the continuous decision layer, a conditional bound for route selection, and feasibility properties for executed actions. The resulting closed loop model provides an empirical and computational basis for studying service restoration across mobility, distribution, communication, and repair operations.

References

- [1] F. He, Y. Yin, and J. Zhou, "Deploying public charging stations for electric vehicles on urban road networks," *Transportation Research Part C: Emerging Technologies*, vol. 60, pp. 227–240, 2015.
- [2] W. Wei, D. Wu, J. Wang, and S. Mei, "Optimal traffic-power flow in urban electrified transportation networks," *IEEE Transactions on Smart Grid*, vol. 8, no. 1, pp. 84–95, 2017.
- [3] K. Qian, C. Zhou, M. Allan, and Y. Yuan, "Coordinated planning of extreme fast charging stations and power distribution networks considering on-site storage," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 1, pp. 493–504, 2021.
- [4] S. Cui, Y. Wang, J. Xiao, and N. Wang, "Optimal pricing of public electric vehicle charging stations considering operations of coupled transportation and power systems," *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3278–3288, 2021.
- [5] K. Li, C. Shao, X. Li, Z. Hu, M. Shahidehpour, and X. Wang, "An inverse optimization method for electric vehicle charging and road toll pricing in coupled urban transportation and power distribution systems," *IEEE Transactions on Smart Grid*, vol. 15, no. 2, pp. 1849–1860, 2024.
- [6] L. Ran, J. Qin, Y. Wan, W. Fu, W. Yu, and F. Xiao, "Fast charging navigation strategy of evs in power-transportation networks: A coupled network weighted pricing perspective," *IEEE Transactions on Smart Grid*, vol. 15, no. 4, pp. 3864–3875, 2024.
- [7] S. M. Rinaldi, J. P. Peerenboom, and T. K. Kelly, "Identifying, understanding, and analyzing critical infrastructure interdependencies," *IEEE Control Systems Magazine*, vol. 21, no. 6, pp. 11–25, 2001.
- [8] M. Ouyang, "Review on modeling and simulation of interdependent critical infrastructure systems," *Reliability Engineering and System Safety*, vol. 121, pp. 43–60, 2014.
- [9] M. Panteli and P. Mancarella, "The grid: Stronger, bigger, smarter? presenting a conceptual framework of power system resilience," *IEEE Power and Energy Magazine*, vol. 13, no. 3, pp. 58–66, 2015.
- [10] Z. Ti, G. Li, M. Zhou, J. Wang, J. Zhao, F. Li, Y. Fu, and X. Wang, "Resilience assessment and improvement for cyber-physical power systems under typhoon disasters," *IEEE Transactions on Smart Grid*, vol. 13, no. 1, pp. 783–794, 2022.
- [11] X. Yin, G. Wu, J. Wei, Y. Shen, H. Qi, and B. Yin, "Deep learning on traffic prediction: Methods, analysis, and future directions," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 6, pp. 4927–4943, 2022.
- [12] W. Jiang and J. Luo, "Graph neural network for traffic forecasting: A survey," *Expert Systems with Applications*, vol. 207, p. 117921, 2022.
- [13] G. Jin, F. Li, J. Zhang, M. Wang, and J. Huang, "Automated dilated spatio-temporal synchronous graph modeling for traffic prediction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 8, pp. 8820–8830, 2023.
- [14] M. Li, Y. Tang, and W. Ma, "Few-sample traffic prediction with graph networks using locale as relational inductive biases," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 8, pp. 8798–8809, 2023.
- [15] Y. Shin and Y. Yoon, "Pgcnn: Progressive graph convolutional networks for spatial-temporal traffic forecasting," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 7, pp. 7633–7644, 2024.

- [16] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital twin in industry: State-of-the-art," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 4, pp. 2405–2415, 2019.
- [17] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108 952–108 971, 2020.
- [18] R. Minerva, G. M. Lee, and N. Crespi, "Digital twin in the iot context: A survey on technical features, scenarios, and architectural models," *Proceedings of the IEEE*, vol. 108, no. 10, pp. 1785–1824, 2020.
- [19] P. Sadeghi-Barzani, A. Rajabi-Ghahnavieh, and H. Kazemi-Karegar, "Optimal fast charging station placing and sizing," *Applied Energy*, vol. 125, pp. 289–299, 2014.
- [20] M. Alizadeh, A. Scaglione, J. Davies, and K. S. Kurani, "A scalable stochastic model for the electricity demand of electric and plug-in hybrid vehicles," *IEEE Transactions on Smart Grid*, vol. 5, no. 2, pp. 848–860, 2014.
- [21] S. Deb, K. Tammi, X. Z. Gao, K. Kalita, P. Mahanta, and S. Cross, "A robust two-stage planning model for the charging station placement problem considering road traffic uncertainty," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 7, pp. 6571–6585, 2022.
- [22] K. Li, C. Shao, Z. Hu, and M. Shahidehpour, "An milp method for optimal planning of electric vehicle charging stations in coordinated urban power and transportation networks," *IEEE Transactions on Power Systems*, vol. 38, no. 6, pp. 5406–5419, 2023.
- [23] S. Lv, X. Zhang, J. Wang, and Y. Liu, "Optimal power and semi-dynamic traffic flow in urban electrified transportation networks," *IEEE Transactions on Smart Grid*, vol. 11, no. 3, pp. 1854–1865, 2020.
- [24] Y. Tao, J. Qiu, S. Lai, X. Sun, H. Liu, and J. Zhao, "Distributed electric vehicle assignment and charging navigation in cyber-physical systems," *IEEE Transactions on Smart Grid*, vol. 15, no. 2, pp. 1861–1875, 2024.
- [25] S. V. Buldyrev, R. Parshani, G. Paul, H. E. Stanley, and S. Havlin, "Catastrophic cascade of failures in interdependent networks," *Nature*, vol. 464, pp. 1025–1028, 2010.
- [26] C. Nowzari, V. M. Preciado, and G. J. Pappas, "Analysis and control of epidemics: A survey of spreading processes on complex networks," *IEEE Control Systems Magazine*, vol. 36, no. 1, pp. 26–46, 2016.
- [27] A. Arab, A. Khodaei, Z. Han, and S. K. Khator, "Proactive recovery of electric power assets for resiliency enhancement," *IEEE Transactions on Smart Grid*, vol. 6, no. 1, pp. 99–109, 2015.
- [28] A. Haydari and Y. Yilmaz, "Deep reinforcement learning for intelligent transportation systems: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 1, pp. 11–32, 2022.
- [29] Z. Zhang, M. Sun, R. Deng, C. Kang, and M.-Y. Chow, "Physics-constrained robustness evaluation of intelligent security assessment for power systems," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 872–884, 2023.
- [30] Z. Zhang, Z. Yang, D. K. Y. Yau, Y. Tian, and J. Ma, "Data security of machine learning applied in low-carbon smart grid: A formal model for the physics-constrained robustness," *Applied Energy*, vol. 347, p. 121405, 2023.
- [31] Z. Zhang, W. Wang, M. Wang, H. Wang, J. Bi, and G. Xu, "Physics-constrained adversarial attack generation and active defense for the hybrid deep reinforcement learning-based load frequency control," *ACM Transactions on Autonomous and Adaptive Systems*, 2026.
- [32] K. Ding, F. T. S. Chan, X. Zhang, G. Zhou, and F. Zhang, "Defining a digital twin-based cyber-physical production system for autonomous manufacturing in smart shop floors," *International Journal of Production Research*, vol. 57, no. 20, pp. 6315–6334, 2019.
- [33] S. Rahmani, A. Baghbani, N. Bouguila, and Z. Patterson, "Graph neural networks for intelligent transportation systems: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 8, pp. 8846–8885, 2023.
- [34] C. Guo, C.-H. Chen, F.-J. Hwang, S.-H. Yang, and C.-C. Chen, "Fast spatiotemporal learning framework for traffic flow forecasting," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 8, pp. 8606–8616, 2023.
- [35] J. Ji, F. Yu, and M. Lei, "Self-supervised spatiotemporal graph neural networks with self-distillation for traffic prediction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 5, pp. 5026–5038, 2023.
- [36] A. Märtz, U. Langenmayr, S. Ried, K. Seddig, and P. Jochem, "Charging behavior of electric vehicles: Temporal clustering based on real-world data," *Energies*, vol. 15, no. 18, p. 6575, 2022.
- [37] C. Brelsford, S. Tennille, A. Myers, S. Chinthavali, V. Tansakul, M. Denman, M. Coletti, J. Grant, S. Lee, K. Allen, E. Johnson, J. Huihui, A. Hamaker, S. Newby, K. Medlen, D. Maguire, C. Dunivan Stahl, J. Moehl, D. Redmon, J. Sanyal, and B. Bhaduri, "A dataset of recorded electricity outages by united states county 2014–2022," *Scientific Data*, vol. 11, no. 1, p. 271, 2024.